

Birational Algebraic Topology

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Abstract: Over a qcqs scheme S , we analyze the birational localization $L_{\text{bir}}\mathcal{H}^{\mathbb{A}^1}(S)$ of the motivic ∞ -category $\mathcal{H}^{\mathbb{A}^1}(S)$. We establish that the associated localization functor L_{bir} commutes with the bar construction, and thus preserves connectivity. Over a perfect field k , we demonstrate that a sheaf of groups is birational exactly when it is strongly $\mathbb{A}^1$ -invariant and has trivial $\mathbb{G}_m$ -contraction. For connected motivic spaces over such fields, this yields a canonical equivalence between L_{bir} and the $S^{2,1}$ -nullification functor $L^{2,1}$ of [ABH23]. Finally, identifying $\pi_0^b(X)$ of a proper scheme X/k with $\pi_0^{b\mathbb{A}^1}(X)$ [AM11], we prove that: 1. the canonical morphism from $\pi_0^{\mathbb{A}^1}(X)$ to $\pi_0^{b\mathbb{A}^1}(X)$ is the universal birationalization, 2. $\pi_0^{b\mathbb{A}^1}(-)$ is a birational invariant of proper schemes, 3. (ind-) proper schemes are $\mathbb{A}^1$ -connected if and only if they are birationally connected.

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1 Introduction to Birational Algebraic Topology

1.1 The aims of birational motivic homotopy theory

Birational geometry studies birational morphisms of schemes. Denoted $X \dashrightarrow Y$, a birational morphism from X to Y stands for a span $X \xleftarrow{i} U \xrightarrow{j} Y$ of dense open immersions (i.e., both i and j have homeomorphic images that are dense in their codomain and induce isomorphisms $i^*\mathcal{O}_X \simeq \mathcal{O}_U \simeq j^*\mathcal{O}_Y$ of structure sheaves). One of the classical questions in birational geometry is the rationality problem: How to determine whether a given variety is birational to $\mathbb{A}^n$? However clean the statement may seem, the rationality problem has no universal solution.

The issue is that birational morphisms are not morphisms in any sufficiently well-behaved 1-category that suitably enlarges the category of (smooth) schemes. Moreover, the rationality question is a rigid one, in the sense that it is really about isomorphism classes. Problems of this type often require one to consider the ‘possible’ $(2, 1)$ -category of such spans, which extends the 1-category of (smooth) schemes in a suitable manner.

One of the best techniques for obtaining negative answers to such problems is to construct invariants that induce isomorphisms on such spans. However, an appropriate, well-behaved invariant must satisfy the properties of a good cohomology theory, for which even 2-categories are insufficient. One shall rather build a category that has all higher categorical colimits. The ‘minimal’ such category, which contains all smooth schemes and is a natural home for such spans, is precisely the point of the birational homotopy category. This is the localization of $\mathcal{P}_\Sigma(Sm_S)$ (see [Lur09, Definition 5.5.8.8] for the notation) at the set of dense open immersions, denoted $\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(Sm_S)$.

Frankly, if one really wants cohomology theories of sufficiently good nature, a priori, it is not clear why our birational category should enforce them. Therefore, it is suggestive to start with a good cohomological extension rather than $\mathcal{P}_\Sigma(Sm_S)$. One of the best and most popular such unstable candidates is, no doubt, the $\mathbb{A}^1$ homotopy category, constructed by Morel and Voevodsky [MV99] (recall that this is defined as the $\mathbb{A}^1$ -localization of the Nisnevich ∞ -topos of Sm_S). So the correct candidate for birational homotopy shall rather be $\mathcal{H}^{b\mathbb{A}^1}(S) := L_{dense}\mathcal{H}^{\mathbb{A}^1}(S)$.

In fact, historically, this was the first construction of the ‘birational motivic homotopy category’ and appeared in [Pel14] for a completely different reason: to study a possible slice filtration of the unstable motivic homotopy category. This unstable theory has a significant impact on motivic slice computations [see BE21]. (We will carry out a similar computation in another work [Mai26d]).

With a different goal in mind, Choudhury and Roy [CR24] considered both constructions over a field k and showed in Theorem 3.4 of *loc.cit.* that they coincide. That is, homotopy invariance and Nisnevich locality are automatic in $\mathcal{H}^b(k)$. This result also appeared in a work of Cisinski and Kahn [CK23, Corollary 2.3]. However, both contain a technical oversight: they rely on the unrestricted ∞ -category of presheaves $\mathcal{P}(Sm_k)$ rather than $\mathcal{P}_\Sigma(Sm_k)$, so that their $\mathbf{H}_b(k)$ (which evaluates to $L_{dense}\mathcal{P}(Sm_k)$ in our setting) contains objects that fail to be local with respect to disjoint covers. The first section of this paper addresses this gap by providing a rigorously corrected version of this strategy while generalizing it to all base schemes. Thus, we show that for any qcqs scheme S , there is an equality of ∞ -categories $\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^b(S)$. (The strategy we develop for automatic Nisnevich locality differs from that of [CR24]. In another work [Mai26c, Theorem 2.3.6], I provide a corrected version of their Corollary 3.7 using the necessary framework of $\mathcal{P}_\Sigma(Sm_S)$, carried out uniformly over all qcqs base schemes.)

Now, obviously, the $\mathbb{A}^1$ homotopy category is not the most general cohomologically rich construction, simply because not all cohomologies are $\mathbb{A}^1$ -invariant. There are a few more on the list that do a little better. Among many such candidates, another unstable cohomological category is the logarithmic motivic homotopy category of [BPOsr22]. In §2.3 of this paper, we will see that even with the logarithmic setup, one does not get anything new: the ‘strict logarithmic birational motivic’ homotopy category is equivalent to the ordinary ‘birational motivic homotopy category’.

We may therefore unambiguously denote the birational motivic homotopy category by $\mathcal{H}^b(S)$. It will therefore follow that computations in $\mathcal{H}^{b\mathbb{A}^1}(S)$ are significantly simpler than their $\mathbb{A}^1$ -homotopical counterparts.

This tractability brings us to the other important aspect: to what extent can birational problems be resolved entirely via $\mathbb{A}^1$ -homotopy theory? This question has been raised and partially answered in various accounts. The starting point of this line of study is probably [AM11], where the authors show that, for smooth proper varieties, the sections of the $\mathbb{A}^1$ -homotopy path component over $\mathcal{F}_k$ capture Manin's notion of R -equivalent points. However, it is known that $\pi_0^{\mathbb{A}^1} X$ (even for a smooth proper variety X) is not birational [BHS15, Example 4.8], so we are forced to consider the 'universal birationalization'. But this birationalization turns out to be extremely difficult to compute by hand.

In later explorations [Aso13], [AH11], the authors have used concepts like this, paired with stability/homology, to detect the existence of rational points or zero cycles of degree one. In fact, they conjectured that the zeroth $\mathbb{A}^1$ homology of such varieties is the free abelian sheaf of their $\pi_0^{b\mathbb{A}^1}$ (this has been verified in [Koi22]) and is itself a birational invariant. Connections of these aspects to birational homotopy theory were established in the other initial references [CR24, Theorem 3.5], [CK23, Proposition 1.9]: the unstable birational path component $\pi_0^b X$ is isomorphic to $\pi_0^{b\mathbb{A}^1} X$ for a smooth proper variety X/k .

In this paper, we remove the smoothness hypothesis via an alternative approach [Theorem 3.6.12] and apply it in Corollary 3.6.15 to show that for a proper scheme X/k , $\pi_0^{b\mathbb{A}^1} X$ [AM11] is precisely the aforementioned universal birationalization of $\pi_0^{\mathbb{A}^1} X$ (in fact, inducing an isomorphism over $\mathcal{F}_k$). Furthermore, this establishes that proper motivic spaces are $\mathbb{A}^1$ -connected if and only if they are birationally connected [Corollary 3.6.13].

It must, however, be noted that the full motivic homotopy type $L_{mot} X$ of a proper scheme X differs significantly from its birational homotopy type $L_{bir} X$. Birational localization thus yields genuinely new birational invariants that remain invisible to the motivic localization.

(In an upcoming work [Mai26b], we shall explain another important aspect of moving from the $\mathbb{A}^1$ -homotopy category to the Birational homotopy category, namely, to add nice topos theoretic properties to $\mathcal{H}^{\mathbb{A}^1}$ (see §1.2 in *loc.cit.*.)

1.2 Notations and terminologies

We shall freely use the language of ∞ -categories without referring to any particular model for the theory. The reader may freely use their preferred model. A standard reference is [Lur09], where the author models these on Kan complexes as ∞ -groupoids. With a choice of Grothendieck universes, the terms Large and small will be treated as usual. The term $\mathrm{Map}_-(-, -)$ will always denote the ∞ -groupoid of morphisms, while $[-, -]_-$ will denote the path component $\pi_0 \mathrm{Map}_-(-, -)$. Spc will stand for the ∞ -category of spaces. Pr^L shall denote the ∞ -category of large presentable ∞ -categories, while Cat_∞ will denote the ∞ -category of all large ∞ -categories.

For an ∞ -category $\mathcal{C}$, by $\mathcal{P}(\mathcal{C})$ we mean the category of presheaves of spaces on $\mathcal{C}$, which is the free co-completion of $\mathcal{C}$. If $\mathcal{C}$ has coproducts, by $\mathcal{P}_\Sigma(\mathcal{C})$ we mean the full subcategory of $\mathcal{P}(\mathcal{C})$ consisting of presheaves that turn finite coproducts into products. This is the sifted cocompletion of $\mathcal{C}$, and therefore the corresponding Yoneda embedding $h : \mathcal{C} \rightarrow \mathcal{P}_\Sigma(\mathcal{C})$ preserves coproducts [Lur09, Proposition 5.5.8.10]. On the other hand, $\mathcal{P}(\mathcal{C})_\bullet$ will denote the full subcategory of $\mathcal{P}(\mathcal{C})$ of pointed objects. pt will stand for the terminal object, so $\mathcal{P}(\mathcal{C})_\bullet = \mathcal{P}(\mathcal{C})_{pt/}$

Throughout this paper, S will denote a scheme, and k will denote a field. In most of the important results, we will need S to be Qcqs and k to be perfect. But we will specify the conditions each time we use them: either at the beginning of the (sub)section or within the statements themselves. We do not assume k is perfect unless otherwise specified. By a morphism of schemes, we shall always mean a separated morphism. Sm_S stands for the category of smooth, finite-type, separated morphisms over S .

We refer to [Hoy14, Appendix C] for the notion of the Nisnevich topology and Nisnevich sheaves on SM_S (smooth schemes over S , not necessarily of finite type) when S is a general scheme. When S is Qcqs, Proposition C.5 (3) of *loc.cit.* yields a canonical equivalence $\mathcal{P}_{nis}(SM_S) \simeq \mathcal{P}_{nis}(Sm_S)$ of the ∞ -toposes of Nisnevich sheaves of spaces. Since we will always work with Qcqs schemes, it is fine to work with Sm_S instead of SM_S .

By $\mathcal{P}(S)$, we mean the ∞ -category of spaces on Sm_S . Throughout, we let σ denote a Grothendieck topology on Sm_S . In most of the important results, we will need σ to be coarser than (but including) the Nisnevich topology. We denote the topos of σ -local (∞ -)sheaves as $\mathcal{P}_\sigma(S)$. The corresponding localization functor, denoted L_σ , is the associated σ (∞)-sheafification and is hence left exact. π_n^σ denotes the homotopy groups internal to the $\mathcal{P}_\sigma(S)$ -local topos, i.e., the σ sheafification of the presheaf π_n .

We shall use the notions of accessible localization and saturation classes throughout the paper. A standard reference is [Lur09, §5.4.4]. We will use the notation $\mathcal{H}^{\mathbb{A}^1}(S)$ for the full subcategory of $\mathcal{P}(S)$ consisting of S -motivic spaces i.e., $\mathbb{A}^1$ -local Nisnevich-local objects. The corresponding localization functors will be denoted $L_{\mathbb{A}^1}, L_{nis}, L_{mot}$, etc. $\pi_i^{\mathbb{A}^1}$ will, as usual, stand for $\pi_i^{nis} L_{mot}$. In principle, one may use a combination of a topology σ and a smooth scheme A/S to construct a localization $\mathcal{H}_\sigma^A(S) := L_A \mathcal{P}_\sigma(S)$.

We use the notation $PSh_S, PGrp_S$, and $PAbs_S$ for the categories of presheaves on Sm_S with values in sets, groups, and abelian groups, respectively. For a topology σ , we use the notation $Sh_S^\sigma, Grp_S^\sigma, Ab_S^\sigma$ for the corresponding subcategories of σ -sheaves. When $\sigma = nis$, we denote these as usual by Sh_S, Grp_S, Ab_S . On the other hand, $Sh_S^{\mathbb{A}^1}, Grp_S^{\mathbb{A}^1}, Ab_S^{\mathbb{A}^1}$ denote the subcategories of ordinary, strongly, and strictly $\mathbb{A}^1$ -invariant sheaves, respectively.

When k is a field, $\mathcal{F}_k$ denotes the category of finitely generated separable extensions of k and field homomorphisms. We write $Sch_k, Prop_k$, and $SmProp_k$ for the categories of all schemes, proper schemes, and smooth proper schemes over k , respectively.

1.3 Summary of the paper

Over a general Qcqs base scheme S , we study the algebraic topology of the Birational Motivic homotopy category $L_{bir}^0 \mathcal{H}^{\mathbb{A}^1}(S)$ of [BE21]. Recall that this is defined as the localization of the Motivic homotopy category $\mathcal{H}^{\mathbb{A}^1}(S)$ at the set $B_0(S)$ of dense open immersions in Sm_S . To shorten notation and be consistent with the notation of [AM11], we denote this localization by $\mathcal{H}^{b\mathbb{A}^1}(S)$. By dropping 0, we denote the localization functor L_{bir}^0 of [BE21] simply by L_{bir} .

The central idea of the paper is that localization at $B_0(S)$ is strong enough to enforce both Nisnevich and $\mathbb{A}^1$ -locality, provided one starts with the sifted cocompletion $\mathcal{P}_\Sigma(Sm_S)$. In fact, it is strong enough to induce locality for any rational scheme.

In other words, if we define:

1. The Birational homotopy category as $\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(Sm_S)$,
2. The Birational Motivic homotopy category as $\mathcal{H}^{b\mathbb{A}^1}(S) := L_{dense}\mathcal{H}^{\mathbb{A}^1}(S)$,
3. The rational motivic homotopy category $\mathcal{H}^{rat}(S)$ as the localization of $\mathcal{P}_{nis}(Sm_S)$ at all rational schemes over S ,

then,

Theorem 1 (see Corollary 2.2.9 and Corollary 2.2.11). $\mathcal{H}^b(S) \subset \mathcal{H}^{rat}(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$. In particular, $\mathcal{H}^{b\mathbb{A}^1}(S) \simeq \mathcal{H}^b(S)$.

In the same section, we define two analogs of the birational localization for the logarithmic motivic homotopy category $\log\mathcal{H}^\square(S)$ [BPOsr22]:

1. $\log\mathcal{H}^{log-b}(S)$: the localization of $\log\mathcal{H}^\square(S)$ at all log-dense open immersions of log schemes,
2. $\log\mathcal{H}^b(S)$: the localization of $\log\mathcal{H}^\square(S)$ at all dense open immersions of log schemes.

We then show that:

Theorem 2 (see Corollary 2.3.6). *There is a canonical equivalence: $\log\mathcal{H}^b(S) \simeq \mathcal{H}^b(S)$ induced by the forgetful functor $\rho : SmlSm_S \rightarrow Sm_S$ given by $(X, D) \mapsto X$.*

By choosing to work with Qcqs schemes, we shall therefore consider $\mathcal{H}^b := L_{dense}\mathcal{P}_\Sigma$ as the candidate for the birational motivic homotopy category. This simplicity of the birational motivic homotopy category will make the corresponding algebraic topology extremely simple. For example, we show that:

Theorem 3 (see Lemma 3.2.3). *L_{bir} commutes with the two-sided (Nisnevich) bar construction.*

One must be cautioned that similar statements are incorrect for the motivic localization functor L_{mot} , because the inclusion $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$ is not closed under geometric realization. The problem stems from a similar issue with the inclusion $\mathcal{P}_{nis}(S) \subset \mathcal{P}(S)$. Although this last inclusion (and hence the other one) is closed under filtered colimits (because of the cd nature of the Nisnevich topology), it is not closed under sifted colimits. This is precisely the reason Morel's notion of Strong/Strict $\mathbb{A}^1$ -invariance arises, which makes ordinary motivic homotopy theory more complicated.

In fact, a higher m connectivity theorem for L_{bir} follows immediately from the same facts (and thus, in fact, over a general base), which, in the case of $\mathcal{H}^{\mathbb{A}^1}$, is highly sensitive to the dimension of the base scheme:

Theorem 4 (see Corollary 3.5.2). *The birational localization functor L_{bir} maps pointed Nisnevich-locally n -connected S -spaces to (sectionwise!) n -connected S -spaces.*

It turns out that commutation of L_{bir} with the bar construction is sufficient to import many more topological results directly from the parent topos. Some of these results also hold in the motivic case, though they require more effort, and having a singular model of $L_{\mathbb{A}^1}$ saves most of them.

Unfortunately, in the birational case, we are unable to construct an explicit formula for L_{bir} in general. However, over perfect fields, we will prove that this is possible, at least for connected spaces. The key observation leading to this model is the following theorem:

Theorem 5 (see Theorem 3.3.12). *Let k be a perfect field. Then the following conditions for a Nisnevich-locally connected pointed motivic space $\mathcal{X}$ over k are equivalent:*

1. $\mathcal{X}$ is birational local.
2. $\pi_i^{nis}\mathcal{X}$ is birational local for all i .
3. $(\pi_i^{nis}\mathcal{X})_{-1} = 0$ for all i .
4. $\mathcal{X}$ is $\mathbb{P}^1$ local (equivalently $S^{2,1}$ null [ABH23]).

In other words, $\tau_{\geq 1}^{nis}\mathcal{H}^b(k) = \tau_{\geq 1}^{nis}(\mathcal{H}^{\mathbb{A}^1}(k) \cap \mathcal{H}^{\mathbb{P}^1}(k))$.

We shall, however, point out that this theorem might fail for non-connected spaces, as Counterexample 3.3.15 shows. Nevertheless, it follows immediately that when k is a perfect field, there is an equivalence of functors

$$L_{bir} \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}(k)} \simeq L^{2,1} \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}(k)}$$

where $L^{2,1}$ denotes the $S^{2,1}$ nullification from [ABH23]. Although the authors of *loc.cit.* provide a formula for $L^{2,1}$, we caution that it is obtained only via a small object argument. To complete this modeling of L_{bir} , we will therefore provide explicit models for $L^{2,1}$ localization using a ‘singular-like’ construction [Theorem 3.4.2].

In the final subsection, we will obtain standard results about connected coverings (see Theorem 3.6.6, Proposition 3.6.7) and then show that:

Theorem 6 (see Corollary 3.6.9). *Suppose $\mathcal{X}$ is a (motivically) pointed $\mathbb{A}^1$ - n -connected S -space. Then the canonical map $\pi_{n+1}^{\mathbb{A}^1}\mathcal{X} \rightarrow \pi_{n+1}^b\mathcal{X}$ is universal among maps to a birational sheaf of sets when $n = -1$; to a birational sheaf of groups when $n = 0$; and to a birational sheaf of abelian groups when $n \geq 1$; i.e., $\pi_{n+1}^b\mathcal{X} \simeq \pi_0^b(\pi_{n+1}^{\mathbb{A}^1}\mathcal{X})$.*

In fact, using Proposition 3.6.7, we will obtain a generalization of [CR24, Theorem 3.5] to all proper schemes (see [AM11, Theorem 6.2.1] for the definition of $\pi_0^{b\mathbb{A}^1}X$):

Theorem 7 (see Theorem 3.6.12, Corollary 3.6.15, and Corollary 3.6.13). *Let k be a field and X a proper scheme over k . Then there is a canonical isomorphism $\pi_0^bX \simeq \pi_0^{b\mathbb{A}^1}X$. It follows that the canonical morphism $\pi_0^{\mathbb{A}^1}X \rightarrow \pi_0^{b\mathbb{A}^1}X$ is the universal birationalization and that $\pi_0^{b\mathbb{A}^1}(-)$ is a birational invariant of proper schemes.*

As a consequence, we obtain:

Theorem 8 (see Corollary 3.6.18). *An (ind-) proper scheme over a field k is $\mathbb{A}^1$ -connected if and only if it is birationally connected.*

As a final application of Proposition 3.6.7, we prove a birational version of the Freudenthal suspension theorem (see Theorem 3.6.22).

We will conclude the paper in §4 with a collection of examples of birational local spaces.

2 Constructing the Birational Motivic Homotopy Category

Let S be a Qcqs scheme. By Sm_S we mean the full subcategory of Sch_S spanned by schemes over S with finitely presented and smooth structure maps. By an S -space we mean a presheaf of spaces on Sm_S . The ∞ -category of S -spaces, denoted $\mathcal{P}(S)$, is defined as the ∞ -category of presheaves of spaces on Sm_S , i.e., $\mathcal{P}(S) := \mathcal{P}(Sm_S) = \text{Fun}(Sm_S^{op}, Spc)$. Since Sm_S is (essentially) small, by [Lur09, Remark 5.5.3.7] the ∞ -category $\mathcal{P}(S)$ is presentable and, in fact, an ∞ -topos [see Lur09, Definition 6.1.0.4]. When the base scheme S is fixed, we refer to S -spaces simply as 'spaces' and implicitly identify any smooth scheme X with its representable presheaf $h_S(X)$.

We say that an open immersion [see Sta26, Definition 01HE] $j : U \hookrightarrow X$ is a dense open immersion if the underlying topological space $j(U)$ is dense in X . We say that a presheaf $\mathcal{X} \in \mathcal{P}(S)$ is dense local if it is local with respect to the class

$$B_0(S) := \{j : U \hookrightarrow X \in Sm_S \mid j \text{ is a dense open immersion}\}$$

of dense open immersions in Sm_S . We denote the full subcategory of $\mathcal{P}(S)$ consisting of dense local spaces by $L_{dense}\mathcal{P}(S)$. Moreover, since $B_0(S)$ is (essentially) small, it follows from [Lur09, Proposition 5.5.4.15] that $L_{dense}\mathcal{P}(S)$ is an accessible localization of $\mathcal{P}(Sm_S)$. Let us denote the associated saturated class and the corresponding localization functor by $Dense(S)$ and L_{dense} , respectively.

It is well known that the Yoneda embedding $h : Sm_S \rightarrow \mathcal{P}(Sm_S)$ does not, in general, preserve coproducts. Consequently, the ∞ -category $\mathcal{P}(S) := \mathcal{P}(Sm_S)$ contains many (geometrically) unidentified homotopy types. As a result, the dense local homotopy category $L_{dense}\mathcal{P}(S)$ is too naive to support meaningful birational geometry. To avoid introducing new homotopy types for existing coproducts, one should replace $\mathcal{P}(Sm_S)$ with the category $\mathcal{P}_\Sigma(Sm_S)$ [Lur09, Definition 5.5.8.8]. This is the full subcategory of $\mathcal{P}(Sm_S)$ consisting of presheaves that send finite coproducts in Sm_S to products in Spc .

Remark 2.0.1. Since Sm_S is an extensive (∞ -)category (i.e., has disjoint coproducts), the category $\mathcal{P}_\Sigma(S) := \mathcal{P}_\Sigma(Sm_S)$ is equivalent to the ∞ -category $\mathcal{P}_\sqcup(Sm_S)$ of sheaves of spaces on Sm_S with the disjoint cover topology (i.e., $\{U_\alpha \rightarrow X\}$ is a cover iff $\sqcup U_\alpha \rightarrow X$ is an isomorphism) [BH21, Lemma 2.4]. Thus, the localization functor $L_\Sigma : \mathcal{P}(Sm_S) \rightarrow \mathcal{P}_\Sigma(Sm_S)$ is left exact and accessible, making $\mathcal{P}_\Sigma(Sm_S)$ an ∞ -topos. Clearly, the Nisnevich topology is finer than the $\sqcup$ -topology and hence $\mathcal{P}_{nis}(S) \subset \mathcal{P}_\Sigma(S)$.

With this understanding, we are now ready to define the correct version of the birational homotopy category:

Definition 2.0.2. A presheaf $\mathcal{X} \in \mathcal{P}(S)$ is said to be *birational local* if it belongs to $\mathcal{P}_\Sigma(S)$ and is dense local. The full subcategory of $\mathcal{P}(S)$ consisting of birational local spaces is called the **Birational Homotopy category**, denoted $\mathcal{H}^b(S)$. Since $B_0(S)$ is an essentially small set, we note that $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is an accessible localization. The associated saturated class is denoted $bir(S)$, and a morphism in it is called a *Birational equivalence*.

In fact, one may strengthen the geometric identification by imposing descent for a finer topology σ and, analogously, define $\mathcal{H}_\sigma^b(S)$ as the localization of $\mathcal{P}_\sigma(Sm_S)$ at the set $B_0(S)$.

Let us denote the corresponding saturated class by $\text{bir}_\sigma(S)$. If σ is finer than τ , then $\mathcal{H}_\sigma^b \subset \mathcal{H}_\tau^b$. In particular, we have

$$\mathcal{H}_{\text{nis}}^b \subset \mathcal{H}_\square^b \equiv \mathcal{H}^b.$$

On the other hand, to obtain a cohomologically rich homotopy category that also respects birationality, one may as well start with the ∞ -category $\mathcal{H}^{\mathbb{A}^1}(S)$ of S -Motivic spaces from [MV99] rather than with $\mathcal{P}_{\text{nis}}(S)$. Recall that a space $\mathcal{X} \in \mathcal{P}(S)$ is an S -Motivic space if it is a nisnevich sheaf and local with respect to the projection maps

$$\mathbb{A}(S) := \{\mathbb{A}_X^1 \rightarrow X\}_{X/S \in \text{Sm}_S}.$$

We define a cohomological analog of the birational category as the localization of the motivic ∞ -category $\mathcal{H}^{\mathbb{A}^1}(S)$ at $B_0(S)$. Let us denote this new category by $\mathcal{H}^{b\mathbb{A}^1}(S)$. Again, since $B_0(S)$ is (essentially) small, this is the full subcategory of $\mathcal{H}^{\mathbb{A}^1}(S)$ consisting of dense local objects, i.e., $\mathcal{H}^{b\mathbb{A}^1}(S) := L_{\text{dense}}\mathcal{H}^{\mathbb{A}^1}(S)$. This is the 0-birational motivic homotopy category $L_{\text{bir}}^0\mathcal{H}^{\mathbb{A}^1}(S)$ of [BE21] (indeed, it is easy to see that 0-birational maps, as defined in *loc.cit.*, are precisely the dense open immersions). We will thus drop the 0 everywhere and call the category $\mathcal{H}^{b\mathbb{A}^1}(S) \equiv L_{\text{bir}}^0\mathcal{H}^{\mathbb{A}^1}(S)$ the Birational **Motivic** homotopy category, and denote the associated localization functor L_{bir}^0 from *loc.cit.* simply by L_{bir} . By construction, there is a canonical natural transformation $L_{\text{mot}} \rightarrow L_{\text{bir}}$ that is an L_{bir} -equivalence.

Since $\mathcal{H}^{\mathbb{A}^1}(S)$ and $\mathcal{P}_\sigma(S)$ are presentable and $B_0(S)$ is essentially small, by [Lur09, Proposition 5.5.4.15], each member of the chain of inclusions

$$\mathcal{H}^{b\mathbb{A}^1} \subset \mathcal{H}_{\text{nis}}^b \subset \mathcal{H}_\square^b$$

is a presentable ∞ -category. The aim of the next two subsections is to show that each of these inclusions is an equality.

2.1 Nisnevich locality of Birational local spaces

In this section, we aim to show that $\mathcal{H}^b(S) \subset \mathcal{P}_{\text{nis}}(S)$. That is, a birational local space, in the sense of Definition 2.0.2, is automatically Nisnevich local. Equivalently, all Nisnevich local equivalences are birational equivalences. In other words,

$$\mathcal{H}^b(S) := L_{\text{dense}}\mathcal{P}_\Sigma(S) = L_{\text{dense}}\mathcal{P}_{\text{nis}}(S) =: \mathcal{H}_{\text{nis}}^b(S).$$

Before we do that, let us note something preliminary:

Proposition 2.1.1. *Let $r : S^{\text{red}} \rightarrow S$ be the reduction map. Then $r^* : \text{Et}_S \rightarrow \text{Et}_{S^{\text{red}}}$ is an equivalence of categories.*

Proof. Since r is a homeomorphism, this follows from [Gro66, Théorème (18.1.2)]. $\square$

Lemma 2.1.2. *Let X/S be smooth. Then, for a dense open immersion $V \hookrightarrow Y$ of schemes smooth over S , the pullback morphism $X \times_S V \rightarrow X \times_S Y$ is also a dense open immersion over S .*

Proof. Openness is standard. For denseness, let $W \subset X \times_S Y$ be an open set. We have to show that $W \cap (X \times_S V) \neq \emptyset$. Since smooth maps are (universally) open, the image $f(W)$ of W under the smooth map $f : X \times_S Y \rightarrow Y$ is open in Y and hence intersects the dense open subset V of Y , say at $y \in V \cap f(W)$. A preimage $t \in W$ of y must therefore be in $X \times_S V$. $\square$

Here is the main theorem of this subsection:

Theorem 2.1.3. *For any Qcqs scheme S , one has $\mathcal{H}^b(S) \subset \mathcal{P}_{nis}(S)$. Therefore, the inclusion $\mathcal{H}_{nis}^b(S) \subset \mathcal{H}_{\square}^b(S) \equiv \mathcal{H}^b(S)$ is an equality.*

Proof. Since our base scheme is Qcqs (i.e., coherent), by [Hoy14, Proposition C.5 (3), (4)], a presheaf on Sm_S is a Nisnevich sheaf if and only if it satisfies Nisnevich excision. By definition, $\mathcal{X} \in \mathcal{P}(Sm_S)$ satisfies Nisnevich excision if it is contractible over the empty scheme and takes Nisnevich cd squares to pullback squares of spaces. So let $\mathcal{X}$ be a birational local space, i.e., $\mathcal{X} \in L_{dense}L_{\Sigma}\mathcal{P}(Sm_S)$. Since $\mathcal{X} \in \mathcal{P}_{\Sigma}(Sm_S)$, we have $\mathcal{X}(\emptyset) \simeq pt$ for free. For the excision part, consider a Nisnevich cd square:

$$\begin{array}{ccc} V & \hookrightarrow & Y \\ q \downarrow & & \downarrow p \\ U & \xhookrightarrow{j} & X \end{array}$$

where j is an open subscheme and p is etale. Consider the open subscheme $W = X \setminus \bar{U}$. Clearly, $U \sqcup W \hookrightarrow X$ is a dense open immersion. Since $p|_{Y \setminus V} : Y \setminus V \rightarrow X \setminus U$ is an isomorphism of reduced schemes and $W \subset X \setminus U$, the pullback $p|_{p^{-1}W}^{red} : p^{-1}W^{red} \rightarrow W^{red}$ is an isomorphism. Since both W and $p^{-1}W$ are etale over X , by Proposition 2.1.1 we deduce that the original map $p|_{p^{-1}W} : p^{-1}W \rightarrow W$ is also an isomorphism. On the other hand, since $U \sqcup W \hookrightarrow X$ is dense, by Lemma 2.1.2 so is $V \sqcup p^{-1}W \hookrightarrow Y$. Now, let us consider the following commutative diagram:

$$\begin{array}{ccccccc} \mathcal{X}(U) \times \mathcal{X}(W) & \xleftarrow{\simeq} & \mathcal{X}(U \sqcup W) & \xleftarrow{\simeq} & \mathcal{X}(X) & \longrightarrow & \mathcal{X}(U) \\ \downarrow q^* \times p^*_{|p^{-1}W} & & & & \downarrow & & \downarrow q^* \\ \mathcal{X}(V) \times \mathcal{X}(p^{-1}W) & \xleftarrow{\simeq} & \mathcal{X}(V \sqcup p^{-1}W) & \xleftarrow{\simeq} & \mathcal{X}(Y) & \longrightarrow & \mathcal{X}(V) \end{array}$$

The horizontal arrows on the left are equivalences because $\mathcal{X} \in L_{\Sigma}\mathcal{P}(Sm_S)$. The horizontal arrows in the middle are equivalences because $\mathcal{X}$ inverts dense open immersions. Since $p|_{p^{-1}W}$ is an isomorphism, the entire outer diagram (with the left and middle horizontal arrows replaced by their inverses) is tautologically a pullback square. Thus, the square on the extreme right (being equivalent to the whole outer diagram) is a pullback square itself. $\square$

Remark 2.1.4. Alternatively, one can directly show that Nisnevich-local equivalences over Qcqs schemes are birational equivalences in our sense. This is established in [Mai26c, Theorem 2.3.4]. The argument there rectifies a slight inaccuracy in the formulation of the analogous result over fields, proven in [CR24, Lemma 3.6]. Because the framework in [CR24] relies on the naive dense local homotopy category (i.e., their birational category $\mathbf{H}_b(k)$ is merely $L_{dense}\mathcal{P}(k)$), the *birational weak equivalences* they consider are, in fact,

just the dense local equivalences in our sense. However, for a nontrivial decomposition $A \sqcup B = X$ (which is a Nisnevich cover) of smooth schemes, the Čech nerve of the presheaf map

$$h_S A \sqcup h_S B \rightarrow h_S X$$

need neither be a sectionwise equivalence nor a dense local equivalence. In [Mai26c, Theorem 2.3.4] we address this discrepancy by providing a modified proof while simultaneously working over an arbitrary Qcqs base scheme.

Remark 2.1.5. One can similarly show that birational motivic spaces are cdh local as well.

We conclude this section with a discussion of the implications of the above theorem. First, we present a generalization of a well-known definition (see [AM11, Definition 6.1.1]):

Definition 2.1.6. A presheaf of sets F on Sm_S is called *dense local* if it is $B_0(S)$ -local in $PSh(S)$. We call it *birational* if, in addition, F lies in $PSh_\Sigma(S)$. In other words, these are precisely the discrete spaces that are birational local.

Remark 2.1.7. From Theorem 2.1.3, it follows that, over any Qcqs scheme S , every birational presheaf is automatically a Nisnevich sheaf. We will denote by Shv_S^b, Grp_S^b , and Ab_S^b the categories of birational (pre)sheaves on Sm_S with values in sets, groups, and abelian groups, respectively.

Proposition 2.1.8. A presheaf $\mathcal{X}$ of spaces is birational-local iff, for every $i \geq 0$, the presheaf $\pi_i \mathcal{X}$ is birational.

Proof. It is clear that $\mathcal{X}$ is dense local iff $\pi_i \mathcal{X}$ is dense local for all i . Thus it suffices to know that $\mathcal{X} \in \mathcal{P}_\Sigma(Sm_S)$ iff $\pi_i \mathcal{X} \in PSh_\Sigma(S)$. This is obvious because π_i preserves finite products. $\square$

It follows that when $\mathcal{X}$ is birational local, the presheaves $\pi_i \mathcal{X}$ are birational presheaves of sets for $i = 0$, of groups for $i = 1$, and of abelian groups for $i \geq 2$. By the remark above, Remark 2.1.7, these are automatically Nisnevich local. Thus, in particular, the birational analog of Morel's $\mathbb{A}^1$ -homotopical result (that higher $\mathbb{A}^1$ -homotopy group sheaves are $\mathbb{A}^1$ -invariant) is straightforward and is true even for the Nisnevich path-component:

Corollary 2.1.9. Let $\mathcal{X}$ be a birational local space over a Qcqs scheme S and let σ be a topology coarser than the Nisnevich topology. Then for all $i \geq 0$, $\pi_i^\sigma \mathcal{X} = \pi_i \mathcal{X}$. It follows that the sheaves $\pi_i^\sigma \mathcal{X}$ are birational local. In particular, the Nisnevich sheaf of sets $\pi_i^{nis} \mathcal{X}$ is birational for all $i \geq 0$.

Proof. By Proposition 2.1.8, we know that $\pi_i \mathcal{X}$ is birational. Using Theorem 2.1.3, we therefore have that $\pi_i \mathcal{X}$ is Nisnevich local and thus σ -local, for all σ coarser than the Nisnevich topology. Consequently, $\pi_i \mathcal{X} = \pi_i^\sigma \mathcal{X}$. Because $\pi_i \mathcal{X}$ is birational, in turn so are $\pi_i^\sigma \mathcal{X}$. $\square$

Remark 2.1.10. An analogous result for $\mathcal{H}^{\mathbb{A}^1}(S)$ is false: π_0^{nis} of an S -motivic space need not be S -motivic. Until the last decade, it was a well-studied conjecture of Morel [Mor12, Conjecture 1.12] and was recently proven to be false by Ayoub [Ayo].

2.2 Rationality of Birational local spaces

The purpose of this section is to show that Birational local spaces are *Rational local*. In particular, $\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$. In fact, we will prove a stronger result that $L_{\text{dense}}\mathcal{P}(Sm_S) \subset L_A\mathcal{P}(Sm_S)$, for any rational scheme $A \in Sm_S$.

Here, by a rational local space, we mean a space that is local with respect to every rational scheme A/S . Recall that a scheme A/S is said to be rational if, for some n , there is a span of dense open immersions $A \hookleftarrow U \hookrightarrow \mathbb{A}_S^n$ over S . The following definition is analogous to the construction of the $\mathbb{A}^1$ -motivic homotopy category:

Definition 2.2.1 (*A-motivic spaces*). *Let $A \in Sm_S$. We say that a presheaf $\mathcal{X}$ of spaces over S is A-local if the canonical map $\mathcal{X}(X) \rightarrow \mathcal{X}(A \times_S X)$ induced by the projection $A \times_S X \rightarrow X$ (where $X \in Sm_S$ is arbitrary) is an equivalence of ∞ -groupoids. Denote by $L_A\mathcal{P}(S)$ the full subcategory of A-local spaces. We say that $\mathcal{X}$ is an A-motivic space if, furthermore, it is Nisnevich local. We denote the full subcategory of A-motivic spaces by $\mathcal{H}^A(S)$.*

Example 2.2.2. The primary example is $\mathcal{H}^{\mathbb{A}^1}(S)$, introduced and studied in [MV99]. For $A = \mathbb{P}_S^1$, this is the projective motivic homotopy category, introduced in [BPOsr22]. We will conduct an extensive study of this second candidate in an upcoming work [Mai26a].

Definition 2.2.3 (*Rational Motivic spaces*). *We say that a presheaf $\mathcal{X}$ of spaces over S is a rational local space (resp. a rational motivic space) if it is an A-local space (resp. A-motivic space) for every rational scheme $A \in Sm_S$. We denote the full subcategory of rational motivic spaces by $\mathcal{H}^{\text{rat}}(S)$.*

Remark 2.2.4. Note that, since Sm_S is (essentially) small, both the inclusions $\mathcal{H}^A(S) \subset \mathcal{P}(S)$ and $\mathcal{H}^{\text{rat}}(S) \subset \mathcal{P}(S)$ are accessible localizations, and the resulting categories of local objects are presentable ∞ -categories [Lur09, Proposition 5.5.4.15]. We call the associated local equivalences A-motivic equivalences and rational motivic equivalences, respectively.

Remark 2.2.5. Rational local spaces are in particular $\mathbb{A}^1$ local, and thus $\mathcal{H}^{\text{rat}}(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$.

Here is the key result relating birationality to rationality:

Lemma 2.2.6. *Let S be a scheme and $A \in Sm_S$ a rational scheme over S . Then a dense local presheaf of sets over S is A-local.*

Proof. Let F be one such presheaf. We first observe that F is $\mathbb{P}^1$ -local, for which one may follow the arguments in [KS15, Appendix A] verbatim. (The assumption in *loc.cit.* that the base is a field is not necessary. See [Mai26c, Lemma 2.3.8] for a quick sketch.)

Since F is dense local, the section at ∞ induces a natural isomorphism $F(\mathbb{P}_S^1 \times_S -) \xrightarrow{\sim} F(\mathbb{A}_S^1 \times_S -)$. The previous paragraph then shows that F is $\mathbb{A}^1$ -local and hence, for all n , $\mathbb{A}^n$ -local as well. Finally, since dense open immersions are stable under base change by smooth morphisms [see Lemma 2.1.2], the span $A \hookleftarrow U \hookrightarrow \mathbb{A}_S^n$ of dense open immersions induces a natural isomorphism $F(A \times_S -) \rightarrow F(\mathbb{A}_S^n \times_S -)$ (again, since F is dense local). By the $\mathbb{A}^n$ -locality of F , this immediately implies the A-locality of F . $\square$

Corollary 2.2.7. *Over a Qcqs scheme, birational presheaves are $\mathbb{A}^1$ -invariant Nisnevich sheaves.*

Proof. By Lemma 2.2.6, such presheaves are $\mathbb{A}^1$ -invariant. These are Nisnevich sheaves by Theorem 2.1.3. $\square$

Lemma 2.2.8. *For any rational scheme A/S we have $L_{dense}\mathcal{P}(S) \subset L_A\mathcal{P}(S)$.*

Proof. A space $\mathcal{E} \in \mathcal{P}(Sm_S)$ is A -local (resp. dense local) if and only if the presheaves $\pi_i(\mathcal{E})$ are A -local (resp. dense local). Therefore, the claim follows immediately from Lemma 2.2.6. $\square$

Corollary 2.2.9. *Let S be a Qcqs scheme. Then:*

1. *For every rational scheme A/S we have,*

$$\mathcal{H}^b(S) = \mathcal{H}_{nis}^b(S) \subset \mathcal{H}^A(S).$$

Equivalently, A -motivic equivalences are birational equivalences.

2. *$\mathcal{H}^b(S) \subset \mathcal{H}^{rat}(S)$. Equivalently, rational motivic equivalences are birational equivalences.*

Proof. The inclusion in (1) follows immediately from Lemma 2.2.8. Indeed, $\mathcal{H}_{nis}^b(S) := \mathcal{P}_{nis}(S) \cap L_{dense}\mathcal{P}(S)$, so that Lemma 2.2.8 yields,

$$\mathcal{H}_{nis}^b(S) \subset \mathcal{P}_{nis}(S) \cap L_A\mathcal{P}(S) = \mathcal{H}^A(S).$$

For the equality in 1, see Theorem 2.1.3.

2. The inclusion in (2) follows immediately from the inclusion in 1 and the definition: $\mathcal{H}^{rat}(S) = \bigcap_{A \in Rat_S} \mathcal{H}^A(S)$ (where Rat_S is the set of all rational schemes in Sm_S).

Finally, the ‘equivalently’ part of each statement is simply the dual of the corresponding inclusion. Indeed, in a presentable ∞ -category $\mathcal{C}$, if S_1 and S_2 are two saturated classes, then $S_1 \subset S_2$ if and only if $L_{S_2}\mathcal{C} \subset L_{S_1}\mathcal{C}$. $\square$

Remark 2.2.10. The birational localization is strictly stronger than the rational localization. Indeed, there are non rational schemes that are birationally contractible. For example, when $S = Speck$ there is a stably rational smooth k -variety that is not rational [AM11, Example 2.6.4]. But stably rational varieties are retract rational, which in turn are birationally contractible.

Corollary 2.2.11. *Let S be a Qcqs scheme. Then,*

1. *$\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$, or, equivalently, motivic equivalences are birational equivalences.*
2. *$\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^b(S)$.*

Proof. statement (1) follows immediately from Corollary 2.2.9(1) applied to $A = \mathbb{A}_S^1$. statement (2) can then be deduced easily from 1. Indeed,

$$\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \cap L_{dense}\mathcal{P}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \cap \mathcal{P}_\Sigma(S) \cap L_{dense}\mathcal{P}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \cap \mathcal{H}^b(S)$$

(the second equality comes from the fact that $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}_{nis}(S) \subset \mathcal{P}_{\Sigma}(S)$). $\square$

Remark 2.2.12. A version of the above corollary appears in [CR24, Theorem 3.9] and [CK23, Theorem 2.3]. In these sources, however, the birational homotopy category of a field k (denoted by $\mathbf{H}_b(k)$ and $\mathcal{H}_b(k)$, respectively) is taken simply to be the naive dense local category $L_{dense}\mathcal{P}(k)$. As detailed in Theorem 2.1.4, this definition is technically incomplete for the purpose of establishing the inclusion $\mathbf{H}_b(S) \subset \mathcal{P}_{nis}(S)$, and thereby the equivalence $\mathbf{H}_b(S) = \mathcal{H}^{b\mathbb{A}^1}(S)$. The obstruction lies in the fact that objects in $L_{dense}\mathcal{P}(k)$ do not generally satisfy descent with respect to Nisnevich covers given by disjoint union decompositions.

2.3 Logarithmic Birational motives

The logarithmic version of motivic homotopy theory has been extensively studied by Binda, Park, and Østvare in a series of papers starting with [BPOsr22], with the goal of representing more cohomology theories. In fact, this approach has been highly successful in representing even topological Hochschild homology, which is known to be $\mathbb{A}^1$ -homotopically contractible.

In the previous subsection, we saw that $\mathcal{H}^b(S)$ is a full subcategory of $\mathcal{H}^{\mathbb{A}^1}(S)$. In fact, using Corollary 2.2.9(1) (applied to the rational schemes $\mathbb{P}^n$), we see that $\mathcal{H}^b(S)$ is also a full subcategory of $\mathcal{H}^{\mathbb{P}^1}(S)$ and $\mathcal{H}^{\mathbb{P}^\bullet}(S)$ [see BPOsr22, Definition 4.0.3, for notation]. This suggests a connection between the birational homotopy category and the logarithmic homotopy category. In this subsection, we will examine this connection. Although the authors of [BPOsr22] define the logarithmic category for fields, we remark that the same can be done over an arbitrary Qcqs scheme. Below, we will thus assume that S is a Qcqs scheme; the reader is free to assume that S is the spectrum of a field.

Consider the forgetful functor $\rho : SmlSm_S \rightarrow Sm_S$ defined by $\rho(\bar{X}, D) := \bar{X}$. This has a canonical right adjoint $\lambda : Sm_S \rightarrow SmlSm_S$ defined by $\lambda(X) := (X, \emptyset)$ [see BPOsr22, §4]. Equipped with the logarithmic Nisnevich topology, ρ is neither a morphism of sites nor does it preserve the interval object, so we cannot obtain motivic left derived functors for ρ (it does work for λ , though).

Theorem 2.3.1. *The induced functor $\rho_{\sharp} : \mathcal{P}(SmlSm_S) \rightarrow \mathcal{P}(S)$ sends log motivic equivalences to birational equivalences. Hence, it induces an adjunction $L_{bir}\rho_{\sharp} : \log\text{-}\mathcal{H}^{\square}(S) \rightleftarrows \mathcal{H}^b(S) : \rho^* \simeq \lambda_{\sharp}$. The right adjoint ρ^* is fully faithful, and we have an embedding $\rho^* : \mathcal{H}^b(S) \hookrightarrow \log\text{-}\mathcal{H}^{\square}(S)$.*

Proof. Local equivalences in $\mathcal{P}(SmlSm_S)$ given by strict Nisnevich covers are sent by $\rho_{\sharp}$ to Nisnevich covers of the total schemes, which are birational equivalences by Theorem 2.1.3. Next, recall that the underlying morphism of total schemes of an admissible blow-up [BPOsr22, Definition 2.5.2, Construction 2.5.9] is given by an ordinary blow-up of schemes. Thus $\rho_{\sharp}$ maps an admissible blow-up to a birational morphism of schemes, which is obviously a birational equivalence. Finally, $\rho(\square \times X) = \mathbb{P}_X^1$, so the projection maps $\square \times X \rightarrow X$ are sent to $\mathbb{P}_X^1 \rightarrow X$, which are birational local equivalences by Corollary 2.2.11.

For the last statement, recall that, by definition of log morphisms, the right adjoint λ is fully faithful at the level of schemes. Therefore, the right derived functor $\lambda_{\sharp} \simeq \rho^*$ is fully

faithful. □

We have thus obtained an equivalence $L_{bir}\rho_{\sharp}^* \rightarrow 1_{\mathcal{H}^b(S)}$. So the image of ρ^* is precisely the locus where the unit $1_{\log-\mathcal{H}^b(S)} \rightarrow \rho^*L_{bir}\rho_{\sharp}$ is an equivalence. We want to describe this image of ρ^* . To do this, we introduce the notion of birational locality in the logarithmic setting.

First, let us call a morphism of log schemes $\underline{U} \rightarrow \underline{X}$ a dense open immersion if the underlying morphism of total schemes $\bar{U} \rightarrow \bar{X}$ is a dense open immersion of schemes and the log structure on $\underline{U}$ is the pullback of that on $\underline{X}$. Denote by $\log-\mathcal{H}^b(S)$ the localization of $\log-\mathcal{H}(S)$ with respect to the class of dense open immersions of log schemes.

Lemma 2.3.2. $\rho_{\sharp} : \mathcal{P}(SmlSm_S) \rightarrow \mathcal{P}(S)$ preserves birational local equivalences. Hence, the adjunction in Theorem 2.3.1 descends to $L_{bir}\rho_{\sharp} : \log-\mathcal{H}^b(S) \rightleftarrows \mathcal{H}^b(S) : \rho^* \simeq \lambda_{\sharp}$.

Proof. This is automatic by the definition of log-dense open immersions and the above theorem, Theorem 2.3.1. □

Lemma 2.3.3. $\lambda_{\sharp} : \mathcal{P}(S) \rightarrow \mathcal{P}(SmlSm_S)$ preserves birational local equivalences. Hence, we have an equivalence $L_{log-bir}\lambda_{\sharp} \simeq \lambda_{\sharp}$. This yields another adjunction $L_{log-bir}\lambda_{\sharp} \simeq \lambda_{\sharp} : \mathcal{H}^b(S) \rightleftarrows \log-\mathcal{H}^b(S) : \lambda^* \simeq \omega_{\sharp}$.

Proof. We know that the class of birational local equivalences is generated by Nisnevich squares and dense open immersions (use Theorem 2.1.3), and that the functor λ trivially preserves Nisnevich squares by the definition of the strict Nisnevich topology on log schemes [BPOsr22, Construction 2.5.9]. It remains to handle dense open immersions, but this is automatic by our definition of log dense open immersions. (Note that, in general, λ does not map $\mathbb{A}^1$ -equivalences to equivalences; but we are bypassing this issue by the result of the previous section that $\mathbb{A}^1$ -equivalences are dense equivalences).

Since, by the lemma above, $\lambda_{\sharp}$ preserves birational local objects, we conclude that $L_{log-bir}\lambda_{\sharp} \simeq \lambda_{\sharp}$ on $\mathcal{H}^b(S)$. □

Corollary 2.3.4. $\rho^* : \mathcal{H}^b(S) \hookrightarrow \log-\mathcal{H}^{\square}(S)$ factors through $\log-\mathcal{H}^b(S)$.

Theorem 2.3.5. The adjunction $\rho^* \simeq \lambda_{\sharp} : \mathcal{H}^b(S) \rightleftarrows \log-\mathcal{H}^b(S) : \lambda^* \simeq \omega_{\sharp}$ as above, is an equivalence.

Proof. As $\lambda_{\sharp}$ is fully faithful, we need to show that $\lambda_{\sharp}\omega_{\sharp} \rightarrow 1$ is an equivalence in $\log-\mathcal{H}^b(S)$. Because both functors have left adjoints even at the birational level, it suffices to check this on generators, namely smooth log schemes over S . Since for any smooth X we know $\lambda\omega(X) = (X \setminus \partial X, \emptyset)$, it suffices to observe that the obvious open immersion $(X \setminus \partial X, \emptyset) \rightarrow (X, \partial X)$ is dense by definition. □

The following is an immediate corollary:

Corollary 2.3.6. A log motivic space $\mathcal{X} \in \log-\mathcal{H}^{\square}(S)$ is in the essential image of ρ^* if and only if it is dense local in $\mathcal{P}(SmlSm_S)$. Thus, ρ^* induces an equivalence $\mathcal{H}^b(S) \simeq \log-\mathcal{H}^b(S)$.

At this point, we caution the reader that there is a weaker notion of log-dense maps than the one introduced above. Namely, a dense open immersion $(U, D_U) \rightarrow (X, D_X)$ of log schemes is called log-dense if, furthermore, $X \setminus D_X \subset U$ (and therefore $X \setminus D_X \subset U \setminus D_U$, since $D_U = D_X \cap U$). Let us denote the localization of $\log\mathcal{H}^\square(S)$ at the log-dense open immersions by $\log\mathcal{H}^{\log-b}(S)$. Since log-dense morphisms are in particular dense, it follows that $\log\mathcal{H}^b(S) \subset \log\mathcal{H}^{\log-b}(S)$. This need not be an equivalence. In fact, this is obtained as a localization at the set $\text{int}(S)$ of interior immersions:

$$\text{int}(S) := \{X^* = (X \setminus D, \emptyset) \rightarrow (X, D)\}_{X \in \text{SmlSm}_S}$$

Proposition 2.3.7. *The inclusion $\log\mathcal{H}^b(S) \subset \log\mathcal{H}^{\log-b}(S)$ is a localization with respect to the set $\text{int}(S)$ of interior immersions. Thus, $\log\mathcal{H}^b(S) = L_{\text{int}}\log\mathcal{H}^{\log-b}(S)$*

Proof. For this, it suffices to observe that a dense open immersion of log schemes is a span of log-dense and interior immersions. Indeed, if $\underline{U} \rightarrow \underline{X}$ is a dense open immersion of log schemes, then in the following commutative square of log schemes:

$$\begin{array}{ccc} U^* & \longleftarrow & X^* \\ \downarrow & & \downarrow \\ \underline{U} & \longrightarrow & \underline{X} \end{array}$$

the vertical maps are interior equivalences, while the top horizontal map $U^* \leftarrow X^*$ is trivially log-dense. By 2-out-of-3 property, $\underline{U} \rightarrow \underline{X}$ is an equivalence in $L_{\text{int}}\log\mathcal{H}^{\log-b}(S)$. $\square$

Actually, it is not difficult to see that $w_\sharp$ induces an equivalence $L_{\text{int}}\log\mathcal{H}^\square(S) \simeq \mathcal{H}^{\mathbb{A}^1}(S)$, and we obtain the following commutative square in Cat_∞ :

$$\begin{array}{ccccc} \mathcal{H}^b(S) & \xrightarrow{\sim} & L_{\text{int}}\log\mathcal{H}^{\log-b}(S) & \hookrightarrow & \log\mathcal{H}^{\log-b}(S) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{H}^{\mathbb{A}^1}(S) & \xrightarrow{\sim} & L_{\text{int}}\log\mathcal{H}^\square(S) & \hookrightarrow & \log\mathcal{H}^\square(S) \end{array}$$

The candidate $\log\mathcal{H}^{\log-b}(S)$ should be regarded as the logarithmically correct candidate for the *logarithmic birational motivic homotopy category* and is canonically larger than $\mathcal{H}^b(S)$. We shall study this candidate in future work.

3 Properties of L_{bir} and the Birational Motivic Homotopy Category

Throughout this section, let S be a Qcqs scheme. The purpose of this section is to study the birational motivic homotopy category $\mathcal{H}^{b\mathbb{A}^1}(S)$ of [BE21, Pel14] and the corresponding localization functor $L_{\text{bir}} : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$. Because this involves localization with respect to two different sets of morphisms, it seems difficult to study. However, as we saw in the previous section, the Birational motivic homotopy category admits a simpler model given by $\mathcal{H}^b(S) := L_{\text{dense}}\mathcal{P}_\Sigma(S)$. Consequently, both $\mathcal{H}^{b\mathbb{A}^1}(S)$ and L_{bir} exhibit surprisingly strong structural properties. For ease of exposition, the subsequent results are formulated in terms of the ∞ -category $\mathcal{H}^b(S)$; however, it should be understood that these statements

fundamentally apply to the more refined birational motivic homotopy category $\mathcal{H}^{b\mathbb{A}^1}(S)$ and are therefore of motivic significance.

While the Nisnevich topology remains our primary focus, we formulate the following results in the general framework of an arbitrary Grothendieck topology σ on Sm_S . Similarly, given a scheme $A \in Sm_S$, we use a general version of $\mathcal{H}^A(S)$, namely $\mathcal{H}_\sigma^A(S) := L_A \mathcal{P}_\sigma(S)$. This recovers $\mathcal{H}^A(S)$ when $\sigma = \text{nis}$ and $\mathcal{P}_\sigma(S)$ when $A = S$. It follows immediately from Theorem 2.1.3 that:

Corollary 3.0.1. *If A is rational, and σ is coarser than (but including) the Nisnevich topology, then $\mathcal{H}^b(S) \subset \mathcal{H}_\sigma^A(S)$. Equivalently, L_σ^A equivalences are birational equivalences.*

3.1 Colimits of birational motives

Theorem 3.1.1. *When σ is a topology coarser than the Nisnevich topology and A/S is a smooth rational scheme over S , the inclusion $\mathcal{H}^b(S) \subset \mathcal{H}_\sigma^A(S)$ is closed under sifted colimits. In particular, the inclusions $\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$ and $\mathcal{H}^b(S) \subset \mathcal{P}_\sigma(S)$ are closed under sifted colimits.*

Proof. Let us first consider the case of the discrete topology with $A = S$, in which case we are dealing with the inclusion $\mathcal{H}^b(S) \subset \mathcal{P}(S)$. Since the dense localization L_{dense} is defined by localizing morphisms of the site Sm_S , we have that $L_{\text{dense}} \mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. Indeed, dense localness is checked on sections, and colimits are computed sectionwise.

Since sifted colimits commute with finite products and presheaves in $\mathcal{P}_\Sigma(S)$ are defined in terms of taking coproducts to products, it is well known that $\mathcal{P}_\Sigma(S) \subset \mathcal{P}(S)$ is closed under sifted colimits. Since $\mathcal{H}^b(S) = \mathcal{P}_\Sigma(S) \cap L_{\text{dense}} \mathcal{P}(S)$, we are done in this case.

The case of a general topology σ , as in the statement, then follows formally. Indeed, consider a diagram $D : K \rightarrow \mathcal{H}^b(S)$ indexed by a sifted simplicial set K . Let $i : \mathcal{H}^b(S) \subset \mathcal{P}(S)$ be the inclusion and consider the colimit $P := \text{colim}_K (i \circ D)$ in $\mathcal{P}(S)$. By the argument sketched above, P is birational local and hence $\text{colim}_K D \simeq P$. To see that this is also the colimit in the σ local category, we compute the colimit in the σ local topos to be

$$\text{colim}_K i_\sigma \circ D = L_\sigma(\text{colim}_K iD) = L_\sigma P.$$

But since P is birational local, it is also Nisnevich local (see Theorem 2.1.3) and hence is σ local whenever σ is coarser than the Nisnevich topology. Therefore, the σ local colimit is identical (in $\mathcal{P}(S)$) to P itself.

The same trick now applies to $\mathcal{H}_\sigma^A(S)$, but with Corollary 3.0.1 instead of Theorem 2.1.3. $\square$

Remark 3.1.2. A similar statement is certainly not true for the ordinary $\mathbb{A}^1$ -motivic localization $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$. Of course, $L_{\mathbb{A}^1} \mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. However, $\mathcal{P}_{\text{nis}}(S) \subset \mathcal{P}(S)$ is not closed under sifted colimits. This is primarily because the Nisnevich sheaf condition is given by taking Nisnevich cd squares to pullback squares, and sifted colimits do not commute with pullbacks. To make up for it, if we Nisnevich sheafify this, $\mathbb{A}^1$ -locality is destroyed. Take, for example, $\mathbb{Z}(\mathbb{G}_m)$, the free abelian sheaf on

Sm_k (where k is a perfect field) generated by $\mathbb{G}_m$ reduced at the unit section. $B(\mathbb{Z}(\mathbb{G}_m))$ is not Nisnevich local but is $\mathbb{A}^1$ -local. The Nisnevich sheafification of this last term is not $\mathbb{A}^1$ -local (i.e., $\mathbb{Z}(\mathbb{G}_m)$ is not strongly $\mathbb{A}^1$ -invariant, see [Cho14, Lemma 5.6]). What commutes with pullbacks, though, is filtered colimits. Thus, $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$ is closed only under filtered colimits, not all sifted colimits.

Lemma 3.1.3. *Over a Qcqs base scheme S , let σ be a topology coarser than the Nisnevich topology yet finer than the disjoint union topology, and let A/S be a rational scheme. Then a model for the birational motivic localization functor $L_{bir} : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ is given by the formula: $L_{bir} \simeq \varinjlim_n (L_\sigma^A L_{dense})^n$. In particular,*

$$L_{bir} \simeq \varinjlim_n (L_\Sigma L_{dense})^n \simeq \varinjlim_n (L_{nis} L_{dense})^n \simeq \varinjlim_n (L_{mot} L_{dense})^n$$

Proof. This is standard. Let $\mathcal{X}$ be an S -space. Consider the colimit of the following diagram:

$$L_{dense}\mathcal{X} \rightarrow L_\sigma^A L_{dense}\mathcal{X} \rightarrow L_{dense} L_\sigma^A L_{dense}\mathcal{X} \rightarrow \cdots \quad (1)$$

The colimit in the statement is the colimit of the even-numbered subdiagram of the above diagram. Since the even-numbered subdiagram is cofinal in the original one, the first colimit in the statement is the same as the colimit of the above diagram 1. Now, the inclusion $\mathcal{P}_\Sigma(S) \subset \mathcal{P}(S)$ is closed under filtered colimits, and a σ sheaf is, in particular, a Σ -presheaf. Therefore, $\varinjlim_n (L_\sigma^A L_{dense})^n$ is a Σ -presheaf. But this colimit is also identical to the colimit of the odd-numbered subdiagram given by

$$L_{dense}\mathcal{X} \rightarrow L_{dense} L_\sigma^A L_{dense}\mathcal{X} \rightarrow \cdots \quad (2)$$

Since each term in this diagram is dense local by construction, so is the colimit. Thus, we obtain that $\varinjlim_n (L_\sigma^A L_{dense})^n \mathcal{X}$ is dense local. Combined with the previous paragraph, this implies that $\varinjlim_n (L_\sigma^A L_{dense})^n \mathcal{X}$ is birational local.

Because L_σ^A equivalences are, in particular, birational equivalences [Corollary 2.2.11], we see that each odd-numbered morphism in diagram 1, is a birational equivalence. On the other hand, the even-numbered morphisms are, by construction, dense equivalences and hence birational. It follows from the stability of the saturation class under colimits that the resulting morphism $\mathcal{X} \rightarrow \varinjlim_n (L_\sigma^A L_{dense})^n \mathcal{X}$ is a birational equivalence. $\square$

Corollary 3.1.4. $L_{bir} : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ preserves sifted colimits.

Proof. Recall that the functor L_Σ preserves sifted colimits [Lur09, Proposition 5.5.8.10(5)]. On the other hand, L_{dense} preserves all colimits. Indeed, $L_{dense} : \mathcal{P}(S) \rightarrow L_{dense}\mathcal{P}(S)$ is a left adjoint and hence preserves all colimits, while $L_{dense}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. The claim then follows from the formula $L_{bir} \simeq \varinjlim_n (L_\Sigma L_{dense})^n$ and the fact that colimits commute with colimits. $\square$

In the next result, for a smooth scheme X/S , we denote by $h_S^b(X)$ the birational localization of the presheaf represented by $X/S \in Sm_S$, i.e., $h_S^b(X) := L_{bir}X = L_{bir}h_S X$.

Lemma 3.1.5. *Let S be Qcqs. For every $X \in Sm_S$, the birational space $h_S^b(X)$ is a compact projective object of $\mathcal{H}^b(S)$.*

Proof. Every smooth X/S is a compact projective object in $\mathcal{P}(Sm_S)$. Since projective objects are defined by preservation (by the corresponding representable functor) of geometric realizations, and compact objects by preservation of filtered colimits, we are done because $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is closed under these colimits by Theorem 3.1.1. $\square$

Lemma 3.1.6. *Let S be a Qcqs scheme. Then, $\mathcal{H}^b(S)$ is a presentable ∞ -category, generated under sifted colimits by the set*

$$\{h_S^b(X) \mid \text{the structure map } p : X \rightarrow S \text{ has property } P\},$$

where P is one of the following properties:

- (a) When S is Qcqs: 1. smooth, 2. proper;
- (b) When S is separated: 3. smooth affine, 4. smooth with affine domain;
- (c) When S is noetherian: 5. projective;
- (d) When S is the spectrum of a field of characteristic zero: 6. smooth projective.

Moreover, these colimits can actually be computed in $\mathcal{P}(S)$ itself.

Proof. By construction, $\mathcal{P}_\Sigma(Sm_S)$ is generated under sifted colimits by smooth X/S . Being its localization, so is $\mathcal{H}^b(S)$. Since by assumption our $X/S \in Sm_S$ are separated of finite type, and S is Qcqs, by Nagata compactification [Sta26, Theorem 0F41] we know that our X/S has compactifications. Thus, by taking the scheme-theoretic image in such a compactification, we obtain a dense $X \subset X'$ such that $X' \rightarrow S$ is proper.

When S is separated, so is every $X \in Sm_S$ by construction. So by choosing Zariski covers (note that since S is Qcqs by Theorem 2.1.3, Zariski equivalences are birational), from 1. we may take the affines as generators.

Now suppose S is Noetherian and X/S is proper by 2. We use [Har77, Exercise 4.10] to find a projective X''/S having a birational morphism $X'' \rightarrow X$ over S .

Finally, by Hironaka's theorem [Ann, Corollary 4.13.], over a field of characteristic 0, every projective variety admits a birational morphism from a smooth projective variety. The 6th class, in this case, is therefore birational to the class in 5.

The last statement follows from Corollary 3.1.4. $\square$

3.2 Preservation of finite limits and the bar construction

Proposition 3.2.1. *The inclusion $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is closed under all limits.*

Proof. This follows from the definition of local objects. $\square$

The first key observation we make in this section is that the birational localization functor L_{bir} is cartesian.

Theorem 3.2.2. *Each possible composition of the functors $\mathcal{P}(S) \xrightarrow{L_\Sigma} \mathcal{P}_\Sigma(S) \xrightarrow{L_{nis}} \mathcal{P}_{nis}(S) \xrightarrow{L_{mot}} \mathcal{H}^{\mathbb{A}^1}(S) \xrightarrow{L_{bir}} \mathcal{H}^b(S)$ is cartesian.*

Proof. Since L_Σ is the sheafification for the disjoint topology and L_{nis} is the sheafification for the Nisnevich topology, both are well known to be left exact. That L_{mot} is cartesian is a standard fact [see, for example, Hoy17, Proposition 3.15].

To conclude, it suffices to show that the composite $L_{bir} : \mathcal{P}(S) \rightarrow \mathcal{H}^b(S)$ is cartesian. Since filtered colimits commute with finite products and L_Σ preserves them, Lemma 3.1.3 immediately reduces the problem to showing that L_{dense} is cartesian.

Suppose X/S is smooth. The cartesian closedness of $\mathcal{P}(S)$ implies that the functor $h_S(X) \times - : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ preserves colimits. Hence, to prove that it preserves $Dense(S)$, it suffices to show that it preserves dense open immersions. This is precisely Lemma 2.1.2. Again, since $\mathcal{P}(S)$ is generated under sifted colimits by smooth X/S and products commute with sifted colimits, we deduce that for any space $\mathcal{X} \in \mathcal{P}(S)$ the product functor $\mathcal{X} \times - : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ preserves dense equivalences. Hence, if $\mathcal{Y}$ is another space in $\mathcal{P}(S)$, the composite

$$\mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{X} \times L_{dense} \mathcal{Y} \rightarrow L_{dense} \mathcal{X} \times L_{dense} \mathcal{Y}$$

is a dense equivalence. But since $L_{dense} \mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under limits, the above composite has a dense local space as its codomain. Hence, $L_{dense} \mathcal{X} \times L_{dense} \mathcal{Y}$ is a dense localization of $\mathcal{X} \times \mathcal{Y}$, meaning that $L_{dense}(\mathcal{X} \times \mathcal{Y}) \rightarrow L_{dense} \mathcal{X} \times L_{dense} \mathcal{Y}$ is an equivalence. In other words, we have proved that L_{dense} preserves binary products.

(Since the terminal object $*$ in $\mathcal{P}(Sm_S)$ is already birational local, it is also the terminal object of $\mathcal{H}^b(S)$, and $L_{bir}(\ast) \simeq \ast$). $\square$

Lemma 3.2.3. *L_{bir} has the following properties:*

1. *L_{bir} takes (group-like) (commutative) monoid S -spaces to (grouplike) (commutative) monoid S -spaces and thus restricts to $L_{bir} : \text{Mon}(\mathcal{P}(Sm_S)) \rightarrow \text{Mon}(\mathcal{H}^b(S))$, $L_{bir} : \text{Mon}(\mathcal{P}(Sm_S))^{gp} \rightarrow \text{Mon}(\mathcal{H}^b(S))^{gp}$, $L_{bir} : \text{CMon}(\mathcal{P}(Sm_S)) \rightarrow \text{CMon}(\mathcal{H}^b(S))$ and $L_{bir} : \text{CMon}(\mathcal{P}(Sm_S))^{gp} \rightarrow \text{CMon}(\mathcal{H}^b(S))^{gp}$. Similarly, L_{bir} preserves monoid actions*
2. *When σ is coarser than the Nisnevich topology, L_{bir} commutes with the two sided σ local bar construction. In fact, $L_{bir} B_\sigma(-, -, -) \simeq B(L_{bir} -, L_{bir} -, L_{bir} -)$. In particular, if $\mathcal{M}$ is a birational monoid that acts on the birational local spaces $\mathcal{X}$ (from the right) and $\mathcal{Y}$ (from the left), then $B_\sigma(\mathcal{X}, \mathcal{M}, \mathcal{Y})$ is birational local.*

Proof. 1. Since L_{bir} is a cartesian localization, statement (1) is obvious.

2. First, assume that σ is the trivial topology. When $\mathcal{M}$ is a monoid acting on $\mathcal{X}$ (from the right) and on $\mathcal{Y}$ (from the left), the first statement implies that $L_{bir} \mathcal{M}$ is a monoid space acting on $L_{bir} \mathcal{X}$ and $L_{bir} \mathcal{Y}$, so $B(L_{bir} \mathcal{X}, L_{bir} \mathcal{M}, L_{bir} \mathcal{Y})$ makes sense. Now, recall that $B(\mathcal{X}, \mathcal{M}, \mathcal{Y}) := |\mathcal{X} \times \mathcal{M}^{\times \bullet} \times \mathcal{Y}|$. Since L_{bir} preserves finite products [see Theorem 3.2.2] and geometric realizations [Corollary 3.1.4], we have

$$\begin{aligned} L_{bir}(B(\mathcal{X}, \mathcal{M}, \mathcal{Y})) &= L_{bir} |\mathcal{X} \times \mathcal{M}^{\times \bullet} \times \mathcal{Y}| \\ &\simeq |L_{bir}(\mathcal{X} \times \mathcal{M}^{\times \bullet} \times \mathcal{Y})| \quad (\text{since } L_{bir} \text{ preserves geometric realization}) \\ &\simeq |L_{bir} \mathcal{X} \times (L_{bir} \mathcal{M})^{\times \bullet} \times L_{bir} \mathcal{Y}| \quad (\text{since } L_{bir} \text{ preserves products}) \\ &= B(L_{bir} \mathcal{X}, L_{bir} \mathcal{M}, L_{bir} \mathcal{Y}) \end{aligned}$$

When σ is a topology coarser than the Nisnevich topology, Theorem 2.1.3 implies that $L_{bir}L_\sigma \simeq L_{bir}$. Therefore,

$$L_{bir}B_\sigma(-, -, -) \simeq L_{bir}L_\sigma B(-, -, -) \simeq L_{bir}B(-, -, -) \simeq B(L_{bir}-, L_{bir}-, L_{bir}-).$$

For the first statement of (2), note that the object $B(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y})$, being birational local already (as it is $\simeq L_{bir}(B(\mathcal{X}, \mathcal{M}, \mathcal{Y}))$), is σ -local by Theorem 2.1.3 and hence

$$B_\sigma(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}) = L_\sigma B(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}) \simeq B(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}).$$

But in the previous paragraph, we have already identified this last term to $L_{bir}B_\sigma(\mathcal{X}, \mathcal{M}, \mathcal{Y})$. $\square$

Corollary 3.2.4. *When σ is coarser than the Nisnevich topology, L_{bir} commutes with the bar construction $B_\sigma := L_\sigma B(*, -, *)$, and thus the restriction functor $L_{bir}B : \text{Mon}(\mathcal{H}^b(S)) \rightarrow \mathcal{H}^b(S)$ coincides with $B_{|\text{Mon}(\mathcal{H}^b(S))}$. In particular, for a (commutative when $n \geq 2$) birational local monoid space $\mathcal{F}$, $B_{nis}^n \mathcal{F}$ is again birational local.*

Remark 3.2.5. This commutation relation is certainly false for L_{mot} and B^{nis} . One reason is that $\mathcal{H}^{mot} \subset \mathcal{P}^{nis}$ is not closed under geometric realizations; see Remark 3.1.2. It is worth noting that the problem has almost nothing to do with the $\mathbb{A}^1$ -localization. In fact, it follows that $BL_{\mathbb{A}^1} \simeq L_{\mathbb{A}^1}B$. To see this, recall that $L_{\mathbb{A}^1}$ commutes with finite products [Hoy17, Corollary 3.5]. Moreover, being a localization at a schematic class, $L_{\mathbb{A}^1} : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ preserves all colimits (indeed, $L_{\mathbb{A}^1} : \mathcal{P}(S) \rightarrow L_{\mathbb{A}^1}\mathcal{P}(S)$ is a left adjoint, while $L_{\mathbb{A}^1}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits). Since the bar construction is given by a geometric realization of products, and both commute with $L_{\mathbb{A}^1}$, we get $BL_{\mathbb{A}^1} \simeq L_{\mathbb{A}^1}B$ just as in Lemma 3.2.3. On the other hand, directed colimits also commute with the bar construction, so in the formulae $\text{colim}_{\rightarrow n} (L_{\mathbb{A}^1} \circ L_{nis})^n$ the only problematic candidate for this desired commutation is the nisnevich localization. Unfortunately, even though L_{nis} preserves products, it fails to preserve geometric realizations. In fact, it fails for any other topology finer than the $\square$ topology.

Corollary 3.2.6. *The canonical functors $K(-, 0)_{|Shv_S^b} = \text{disc} : Shv_S^b \rightarrow \mathcal{P}_\Sigma(Sm_S)$, $K(-, 1)_{|Grp_S^b} = B_{|Grp_S^b} : Grp_S^b \rightarrow \mathcal{P}_\Sigma(Sm_S)$ and $K_{|Ab_S^b}(-, n) = B_{|Ab_S^b} : Ab_S^b \rightarrow \mathcal{P}_\Sigma(Sm_S)$ factors through the birational local subcategory $\mathcal{H}^b(S)$ and are identical to $K^\sigma(-, n)$ for any σ coarser than the Nisnevich topology.*

Corollary 3.2.7. *Birational (commutative) monoid spaces are (strictly) strongly $\mathbb{A}^1$ -invariant. i.e., $\text{Mon}(\mathcal{H}^b) \subset \text{Mon}_{mot}(\mathcal{H}^{\mathbb{A}^1})$ and $\text{CMon}(\mathcal{H}^b) \subset \text{CMon}_{mot}(\mathcal{H}^{\mathbb{A}^1})$. In particular, $Grp_S^b \subset Grp_S^{\mathbb{A}^1}$, $Ab_S^b \subset Ab_S^{\mathbb{A}^1}$, where $Grp_S^{\mathbb{A}^1}$ (resp. $Ab_S^{\mathbb{A}^1}$) stand for strongly (resp. strictly) $\mathbb{A}^1$ -invariant Nisnevich sheaves of groups (resp. abelian groups).*

Proof. Since birational local spaces are $\mathbb{A}^1$ -local, the result follows from Lemma 3.2.3. For example, let $\mathcal{M}$ be a birational monoid. By Corollary 3.2.4, $B_{nis}\mathcal{M}$ is birational local. By Corollary 2.2.11, it follows that $B_{nis}\mathcal{M}$ is motivic local. This is precisely the definition of strong $\mathbb{A}^1$ invariance of a monoid space [EHK⁺21, Definition 3.1.6]. $\square$

Remark 3.2.8. This strengthens Remark 2.3 in [BHS23]. In fact, our argument is entirely different from that in *loc.cit.* and works over all Qcqs base schemes.

Therefore, surprisingly, we see that defining strongly/strictly birational group sheaves (just like strongly/strictly $\mathbb{A}^1$ -invariant group sheaves) is unnecessary.

Corollary 3.2.9. *For σ coarser than the Nisnevich topology, a σ -local presheaf of grouplike monoids $\mathcal{G}$ is birational if and only if $B_\sigma \mathcal{G}$ is birationally local.*

Proof. The only-if part is just Corollary 3.2.4. For the if part, it suffices to observe that, because $\mathcal{G}$ is grouplike, $\mathcal{G} \simeq \Omega B_\sigma \mathcal{G}$, and that Ω preserves local objects. $\square$

Corollary 3.2.10. *The inclusion $Ab_S^b \subset Ab_S^\sigma$ is closed under all colimits and limits and, in particular, is exact. (See [Mai26d, Corollary 3.2.3] for a different argument.)*

Proof. Clearly, the inclusion is closed under all limits. Let us now show the right continuity of the inclusion. First, $L_{dense} Ab_S \subset Ab_S$ is closed under all colimits. By definition, $Ab_S^b = L_{dense} Ab_S \cap L_\Sigma Ab_S$. It thus suffices to show that $L_\Sigma Ab_S \subset Ab_S$ is also closed under all colimits. Since Ab_S is semiadditive, i.e., products are coproducts, this follows from the definition of L_Σ and the fact that taking the binary direct sum of abelian groups commutes with all colimits. $\square$

Lemma 3.2.11. *When σ is coarser than the Nisnevich topology, L_{bir} preserves fiber sequences $\Gamma \rightarrow \mathcal{X} \rightarrow \mathcal{Y}$ of σ -local spaces with σ -locally connected base $\mathcal{Y}$.*

Proof. Every such fiber sequence is equivalent to a principal fiber sequence [AWW17, Theorem 2.3.3 Step 1, Step 2] by realizing $\mathcal{X} \simeq \mathbf{B}_\sigma(\Gamma, \Omega \mathcal{Y}, pt)$ and $\mathcal{Y} \simeq \mathbf{B}_\sigma \Omega \mathcal{Y}$.

$$\begin{array}{ccccc} \Gamma & \longrightarrow & \mathcal{X} & \longrightarrow & \mathcal{Y} \\ \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \Gamma & \longrightarrow & \mathbf{B}_\sigma(\Gamma, \Omega \mathcal{Y}, pt) & \longrightarrow & \mathbf{B}_\sigma \Omega \mathcal{Y} \end{array}$$

Now we apply L_{bir} to the above diagram. Using Lemma 3.2.3 (2), the resulting diagram looks like:

$$\begin{array}{ccccc} L_{bir} \Gamma & \longrightarrow & L_{bir} \mathcal{X} & \longrightarrow & L_{bir} \mathcal{Y} \\ \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ L_{bir} \Gamma & \longrightarrow & \mathbf{B}_\sigma(L_{bir} \Gamma, L_{bir} \Omega \mathcal{Y}, pt) & \longrightarrow & \mathbf{B}_\sigma L_{bir} \Omega \mathcal{Y} \end{array}$$

But the bottom horizontal sequence $L_{bir} \Gamma \rightarrow \mathbf{B}(L_{bir} \Gamma, L_{bir} \Omega \mathcal{Y}, pt) \rightarrow \mathbf{B} L_{bir} \Omega \mathcal{Y}$ is clearly a principal fiber sequence for the action of the grouplike space $L_{bir} \Omega \mathcal{Y}$ on $L_{bir} \Gamma$.

It follows that the top horizontal sequence $L_{bir} \Gamma \rightarrow L_{bir} \mathcal{X} \rightarrow L_{bir} \mathcal{Y}$, being equivalent to the bottom one, is a fiber sequence as promised. $\square$

Theorem 3.2.12. *When σ is coarser than the Nisnevich topology, the localization functor L_{bir} commutes with Ω on $\tau_{\geq 1}^\sigma \mathcal{P}_\sigma(Sm_S)_\bullet$.*

Proof. Apply the above lemma to the fiber sequence $\Omega \mathcal{E} \rightarrow pt \rightarrow \mathcal{E}$. $\square$

Let us remind the reader that the motivic localization, even though not left exact, has a very nice property beyond cartesianness that makes certain motivic computations work well within $\mathcal{P}_{nis}(S)$ without taking motivic localizations. This is the property of local cartesianness of L_{mot} , which is the correct ∞ -categorical analog of a right proper Bousfield localization.

Unfortunately, because dense open immersions are not stable under arbitrary pullbacks, L_{bir} is not expected to be locally cartesian. The following and its corollary are a weaker version of this property. We record them here for completeness and for future use in an upcoming work [Mai26b, §4.4]. The proof below requires a birational connectivity result for L_{bir} , which is an immediate consequence of Lemma 3.2.3 and will be stated formally in Corollary 3.5.2.

Lemma 3.2.13. *L_{bir} is locally cartesian on σ -locally connected pointed objects. That is, if $\mathcal{X}$ is a σ -locally connected space, the morphism $\mathcal{Y} \rightarrow \mathcal{W}$ is in $\mathcal{H}^b(S)$, and $\mathcal{X} \rightarrow \mathcal{W}$ is given, then the morphism $L_{bir}(\mathcal{X} \times_{\mathcal{W}} \mathcal{Y}) \rightarrow L_{bir}(\mathcal{X}) \times_{\mathcal{W}} \mathcal{Y}$ is an equivalence.*

Proof. By factoring $\mathcal{X} \rightarrow \mathcal{W}$ through $L_{bir}\mathcal{X}$, we may assume that $\mathcal{X} \rightarrow \mathcal{W}$ is a birational weak equivalence, and thus $L_{bir}\mathcal{X} \simeq \mathcal{W}$. Let $\mathcal{Z} := \mathcal{X} \times_{\mathcal{W}} \mathcal{Y}$. We have to show that the canonical map $L_{bir}\mathcal{Z} \rightarrow L_{bir}\mathcal{X} \times_{L_{bir}\mathcal{X}} \mathcal{Y} \simeq \mathcal{Y}$ is an equivalence. Let us consider the evident morphism of fiber sequences on the left and their birationalizations on the right:

$$\begin{array}{ccccc} \Gamma & \longrightarrow & \mathcal{Z} & \longrightarrow & \mathcal{X} \\ \phi \downarrow & & \downarrow & & \downarrow \\ \Lambda & \longrightarrow & \mathcal{Y} & \longrightarrow & \mathcal{W} \end{array} \qquad \begin{array}{ccccc} L_{bir}\Gamma & \longrightarrow & L_{bir}\mathcal{Z} & \longrightarrow & L_{bir}\mathcal{X} \\ L_{bir}\phi \downarrow & & \downarrow & & \downarrow \\ \Lambda & \longrightarrow & \mathcal{Y} & \longrightarrow & \mathcal{W} \end{array}$$

Since ϕ is an equivalence in $\mathcal{P}(S)$ (for example, by the pasting lemma for pullbacks) and Λ is birational local, we conclude that $L_{bir}(\Gamma) \simeq \Lambda$ (in fact, $\Gamma \simeq L_{bir}\Gamma$). Because L_{bir} preserves σ -locally connected fiber sequences [Lemma 3.2.11], the top line of the diagram on the right is a fiber sequence. The left-hand square in the same diagram (on the right) is a pullback square since the third leg $L_{bir}\mathcal{X} \rightarrow \mathcal{W}$ is an equivalence and the whole diagram is a morphism of fiber sequences. Moreover, the left bottom map $\Lambda \rightarrow \mathcal{Y}$ in that diagram is an effective epimorphism (since $\mathcal{W}$ is equivalent to $L_{bir}\mathcal{X}$, which is connected by birational connectivity Corollary 3.5.2). By [Lur09, Lemma 6.2.3.16], it is evident that the middle map $L_{bir}\mathcal{Z} \rightarrow \mathcal{Y}$ is an equivalence. This is exactly what we sought to prove. $\square$

Corollary 3.2.14. *L_{bir} is locally cartesian on pointed $\mathbb{A}^1$ -connected objects.*

Proof. Since L_{mot} is locally cartesian [Hoy17, Proposition 3.15] and $L_{bir}L_{mot} \simeq L_{bir}$ [see Corollary 2.2.11], we may apply L_{mot} to the whole setup and assume that $\mathcal{X}$ is Nisnevich connected. $\square$

Remark 3.2.15. L_{bir} is not left exact. For example, let $E \subset X$ be the closed embedding of a birational local curve (e.g., an elliptic curve in $\mathbb{P}^2$; see Theorem 4.1.2) with open complement $U \hookrightarrow X$. Since $U \hookrightarrow X$ is dense, we have

$$L_{bir}(E \times_X U) = L_{bir}(\emptyset) \simeq \emptyset \neq E = L_{bir}E = L_{bir}E \times_{L_{bir}X} L_{bir}U.$$

3.3 Birationality vs Null motivic spaces

In Lemma 3.1.3, we have constructed a standard formula for birational localization. However, without an explicit model for dense localization, the referred-to model is insufficient for meaningful computations. The aim of the next section is to show that when $\mathcal{X}$ is $\mathbb{A}^1$ connected and the base is a perfect field, there is an explicit model for $L_{bir}\mathcal{X}$, given by motivic $S^{1,1}$ and, equivalently, $S^{2,1}$ localization. (This idea of localizing the motivic

homotopy category at motivic spheres has been studied extensively in [ABH23 §3] under the name ‘null motivic spaces’.) To establish the connection between birational local spaces and $S^{2,1}$ (or $S^{1,1}$) null motivic spaces, we will need a few preliminary observations, which we present below.

Lemma 3.3.1. *Suppose $\Gamma \rightarrow \mathcal{X} \rightarrow \mathcal{Y}$ is a fiber sequence in $\mathcal{P}_\sigma(S)$, with $\mathcal{Y}$ a pointed σ (coarser than Nisnevich)-locally connected space. If Γ and $\mathcal{Y}$ are birational local, then so is $\mathcal{X}$.*

Proof. As in the proof of Lemma 3.2.11, we know that $\mathcal{X} \simeq B_\sigma(\Gamma, \Omega\mathcal{Y}, pt)$, which is birational local by Lemma 3.2.3 (2), since both Γ and $\Omega\mathcal{Y}$ are birational local. $\square$

Remark 3.3.2. When σ is non-trivial (for example, the Nisnevich topology), the standard method for proving such statements (see, for example, [Mor12, Lemma 6.51]) is not applicable here. Indeed, when the method in *loc.cit.* is adapted to the birational setup, it requires one to trace the following map of fiber sequences of Nisnevich-local spaces, induced by a non-trivial dense open immersion $U \hookrightarrow X$:

$$\begin{array}{ccccc} \Gamma^X & \longrightarrow & \mathcal{X}^X & \longrightarrow & \mathcal{Y}^X \\ \downarrow j_\Gamma^* & & \downarrow j_\mathcal{X}^* & & \downarrow j_\mathcal{Y}^* \\ \Gamma^U & \longrightarrow & \mathcal{X}^U & \longrightarrow & \mathcal{Y}^U \end{array}$$

But the problem is that even if $\mathcal{Y}$ is Nisnevich connected, there is no reason for either $\mathcal{Y}^X$ or $\mathcal{Y}^U$ to be so. So the morphism $\Gamma^U \rightarrow \mathcal{X}^U$ need not be an effective epimorphism.

Remark 3.3.3. There are non-birational presheaves of sets in $PSh_\Sigma(Sm_S)$ (and even in $PSh_\Sigma^{\mathbb{A}^1}(Sm_S)$) whose Nisnevich sheafification is birational. A general class of such examples is as follows: take a non-birational sheaf of groups G in $Grp_k^{\mathbb{A}^1}$ (say, $\mathbb{G}_m$). Let $F := H_{nis}^1(-; G)$ (for $\mathbb{G}_m$ this is the Picard group functor). Since G is strongly $\mathbb{A}^1$ -invariant, we have $F(\mathbb{A}_X^1) = F(X)$, so $F \in PSh_\Sigma^{\mathbb{A}^1}(Sm_S)$. Moreover, its Nisnevich sheafification $L_{nis}F$ is trivial, hence birational. But clearly, F itself is not birational:

$$F(\mathbb{P}_k^1) = [\Sigma\mathbb{G}_m, \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}} = [\mathbb{G}_m, G]_{\mathcal{H}^{\mathbb{A}^1}} = G(\mathbb{G}_m) = X^*G \neq pt = F(\mathbb{A}_k^1).$$

This kind of failure is also present in the case of $\mathbb{A}^1$ -localization. Namely, there are non- $\mathbb{A}^1$ -invariant presheaves of sets in $PSh_\Sigma(Sm_S)$ whose Nisnevich sheafification is $\mathbb{A}^1$ -invariant. Here is an example. As above, take a non- $\mathbb{A}^1$ -local Nisnevich sheaf G of groups in $PSh_\Sigma(Sm_S)$ (say, $\mathbb{G}_a$) over a proper scheme S . The presheaf $F := H_{zar}^1(-, \mathbb{G}_a)$ has trivial Nisnevich sheafification (since Zariski torsors are Nisnevich torsors as well), implying that F^{nis} is $\mathbb{A}^1$ -local. But since $\mathbb{A}_{\mathbb{Z}}^1$ is affine, by Serre vanishing and the Kunneth formulae, one has $F(\mathbb{A}_S^1) = F(S) \otimes H_{zar}^0(\mathbb{A}_{\mathbb{Z}}^1, \mathbb{G}_a) = F(S)[t] \neq F(S)$.

Lemma 3.3.4. *Suppose S is a Qcqs scheme. For a space $\mathcal{X} \in \mathcal{P}(S)$, the following conditions are equivalent.*

1. $\mathcal{X}$ is birational local.
2. $\pi_i\mathcal{X}$ is birational local for all $i \geq 0$.
3. $\tau_{\leq i}\mathcal{X}$ is birational local for all $i \geq 0$.

Proof. The equivalence $1 \iff 2$ is Proposition 2.1.8. Since $\pi_j \mathcal{X} = \pi_j \tau_{\leq i} \mathcal{X}$, for all $j \leq i$, the equivalence $3 \iff 2$ follows from the case of $1 \iff 2$. $\square$

Proposition 3.3.5. *Suppose S is a Qcqs scheme and σ is a topology coarser than the Nisnevich topology. For a space $\mathcal{X} \in \mathcal{P}_\sigma(S)$, consider the following conditions:*

1. $\mathcal{X}$ is birational local.
2. $\tau_{\leq i}^\sigma \mathcal{X}$ is birational local for all $i \geq 0$.
3. $\pi_i^\sigma \mathcal{X}$ is a birational presheaf for all $i \geq 0$.

Then:

- (a) $1 \implies 2 \implies 3$.
- (b) If $\mathcal{X} \in \tau_{\geq 1}^\sigma \mathcal{P}_\sigma(S)_\bullet$, then $3 \implies 2$.
- (c) If σ is postnikov complete then $2 \implies 1$.

In particular, when $\sigma = \text{nis}$, S is a Qcqs scheme of finite Krull dimension, and $\mathcal{X}$ is pointed and Nisnevich connected, the three conditions 1, 2, and 3 are equivalent.

Proof. (a). Assume 1 holds. Then, by Lemma 3.3.4 we know that $\tau_{\leq i} \mathcal{X}$ is birational local. Since birational local spaces are automatically Nisnevich local [Theorem 2.1.3] it follows that $\tau_{\leq i} \mathcal{X}$ is Nisnevich local. And hence $\tau_{\leq i} \mathcal{X} \simeq L_\sigma \tau_{\leq i} \mathcal{X} =: \tau_{\leq i}^\sigma \mathcal{X}$. Since $\tau_{\leq i} \mathcal{X}$ is birational local, it follows that so is $\tau_{\leq i}^\sigma \mathcal{X}$. That is 2 holds.

Performing the same line of arguments with $\pi_i \mathcal{X}$ instead of $\tau_{\leq i} \mathcal{X}$ we see that $1 \implies 3$. We may now apply this $1 \implies 3$ result to the sheaves $\mathcal{Y}_i := \tau_{\leq i}^\sigma \mathcal{X}$ and obtain $2 \implies 3$.

(b). For the implications $3 \implies 2$ we use standard ideas of Postnikov tower. So assume 3 holds. First, consider the first slice of the Postnikov tower in the topos $\mathcal{P}_\sigma(S)$:

$$L_\sigma K(\pi_1^\sigma(\mathcal{X}), 1) \rightarrow \tau_{\leq 1}^\sigma \mathcal{X} \rightarrow \tau_{\leq 0}^\sigma \mathcal{X} = pt$$

Note that, since $\pi_1^\sigma \mathcal{X}$ is given to be birational, by Lemma 3.2.6 we know that $K^\sigma(\pi_1^\sigma(\mathcal{X}), 1)$ is birational local. Thus, the fiber in the above sequence is birational local. Since the base $\tau_{\leq 0}^\sigma \mathcal{X}$ is contractible (because $\mathcal{X}$ is σ -locally connected), it follows that $L_\sigma K(\pi_1^\sigma(\mathcal{X}), 1) \simeq \tau_{\leq 1}^\sigma \mathcal{X}$, and therefore $\tau_{\leq 1}^\sigma \mathcal{X}$ is birational local. Inductively, assume that $\tau_{\leq n}^\sigma \mathcal{X}$ is birational local. We have the following fiber sequence:

$$K^\sigma(\pi_{n+1}^\sigma(\mathcal{X}), n+1) \rightarrow \tau_{\leq n+1}^\sigma \mathcal{X} \rightarrow \tau_{\leq n}^\sigma \mathcal{X}$$

Since $\mathcal{X}$ is σ -locally connected, we know that so is $\tau_{\leq n}^\sigma \mathcal{X}$. On the other hand, since $\pi_{n+1}^\sigma \mathcal{X}$ is given to be birational local, by Lemma 3.2.6 we deduce that the fiber $K^\sigma(\pi_{n+1}^\sigma(\mathcal{X}), n+1)$ is birational local. Thus, in the last fiber sequence, both the base and the fiber satisfy the conditions of Lemma 3.3.1, and we obtain the birational locality of $\tau_{\leq n+1}^\sigma \mathcal{X}$. By induction, we conclude that all the truncations $\tau_{\leq n}^\sigma \mathcal{X}$ are birational local. This proves $3 \implies 2$.

(c). It remains to prove that $3 \implies 1$ under the assumption that $\mathcal{P}_\sigma(S)$ is postnikov complete. To see how this works, assume all truncations $\tau_{\leq n}^\sigma \mathcal{X}$ are birational local. By Postnikov completeness of $\mathcal{P}_\sigma(S)$, we have $\mathcal{X} \simeq \varprojlim_n \tau_{\leq n}^\sigma \mathcal{X}$. Since the inclusion $\mathcal{H}^b(S) \subset \mathcal{P}_\sigma(S)$ is closed under limits [Proposition 3.2.1], it follows that $\varprojlim_n \tau_{\leq n}^\sigma \mathcal{X} \in \mathcal{H}^b(S)$, and consequently, $\mathcal{X} \in \mathcal{H}^b(S)$.

For the ‘in particular’ part, use the well-known fact that when S is Qcqs of finite Krull dimension, the topos $\mathcal{P}_{nis}(S)$ is Postnikov complete. To see this, use [CM21, Theorem 3.18] and [Lur09, Proposition 7.2.1.10]. $\square$

Remark 3.3.6. For the proof of (b), however, the methods of Remark 3.3.2 do work. The reason is that in this case $(\tau_{\leq n}^{nis}\mathcal{X})^X$ is connected. In fact, when $\tau_{\leq n}^{nis}\mathcal{X}$ is birational local, then so is $\pi_n^{nis}\tau_{\leq n}^{nis}\mathcal{X}$ [see Corollary 2.1.9]. This implies, in particular, that this last term is birational local and hence flasque. Therefore, over a Noetherian base, it follows that $\pi_0(\tau_{\leq n}^{nis}\mathcal{X})^X \simeq H_{nis}^n(X \times -, \pi_n^{nis}\mathcal{X})$ is trivial for all $n \geq 1$ [see Rio02, Lemma 1.39 and 1.40], so that $\pi_0^{nis}(\tau_{\leq n}^{nis}\mathcal{X})^X = *$.

Birationality and contraction

Let G be a presheaf of groups on a Qcqs scheme S . The $\mathbb{G}_m$ -contraction of G is defined as

$$G_{-1} := \Omega_{\mathbb{G}_m^{\wedge 1}} G = \ker\left(G(\mathbb{G}_{m,S} \times_S -) \xrightarrow{ev_1} G\right),$$

where the map $G(\mathbb{G}_{m,S} \times_S -) \xrightarrow{ev_1} G$ is the restriction along the unit section $1 : S \rightarrow \mathbb{G}_{m,S}$.

Lemma 3.3.7. *If G is a dense local presheaf of groups over S , then $G_{-1} = 0$.*

Proof. By Lemma 2.2.6, G is $\mathbb{G}_m$ -local. That is $G(\mathbb{G}_{m,S} \times_S -) \rightarrow G$ is an isomorphism. Since G_{-1} is the kernel of this map, it is trivial. $\square$

Remark 3.3.8. In the rest of the section, we will assume that the base is a perfect field. The reason is mostly generic smoothness. In particular, given a dense open immersion $U \hookrightarrow X$ one can find a filtration of X by open subschemes $U = U_n \subset U_{n-1} \subset \cdots \subset U_1 \subset U_0 = X$ such that for all i , $U_i \setminus U_{i-1} =: C_i$ is smooth and has trivial normal bundle in U_i [see Remark 5.2 of Lev10]. In fact, if $\text{cod}_X(X \setminus U) \geq d$, then it is clear that $\text{cod}_{U_i}(C_i) \geq d$ for all i . Thus, given a presheaf, any problem related to its injectivity or surjectivity of its restriction maps along open immersions can be reduced to checking the same on open immersions with smooth complement having a trivial normal bundle.

Lemma 3.3.9. *Suppose G is a presheaf of (not necessarily abelian) groups over a perfect field k . Then G is birational if and only if it is a strongly $\mathbb{A}^1$ -invariant Nisnevich sheaf with $G_{-1} = 0$.*

Proof. The only if part follows from Corollary 3.2.7 and Lemma 3.3.7.

Conversely, let G be a strongly $\mathbb{A}^1$ -invariant Nisnevich sheaf of groups with $G_{-1} = 1$. Since such sheaves are weakly unramified [Bac24, Corollary 2.4], i.e., are injective on dense open immersions, to show that G is birational, it suffices to show that the restriction map $G(X) \rightarrow G(U)$ is surjective for a dense open immersion $U \hookrightarrow X$. Moreover, by Remark 3.3.8, we may assume that the complement $Z = X \setminus U$ is smooth and has a trivial normal bundle in X . In this situation, purity gives a cofiber sequence $Th_Z(N_Z X) \rightarrow \Sigma U \rightarrow \Sigma X$, where $N_Z X$ is the normal bundle of Z in X . Mapping this sequence to $\mathbf{B}^{nis}G$ (which is given to be $\mathbb{A}^1$ -local) in $\mathcal{H}^{\mathbb{A}^1}(k)$, we are left to show that

$$[Th_Z(N_Z X), \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}(k)}$$

is trivial. Since $U \hookrightarrow X$ is dense, the normal bundle has positive rank, say $n \geq 1$. Because Z has trivial normal bundle in X , i.e., there is an isomorphism $N_Z X \simeq Z \times \mathbb{A}_k^n$, we have

$$Th_Z(N_Z X) \simeq Z_+ \wedge T^n \simeq^{mot} \Sigma^n Z_+ \wedge \mathbb{G}^{\wedge n}.$$

So we see that indeed,

$$\begin{aligned} [Th_Z(N_Z X), \mathbf{B}^{nis} G]_{\mathcal{H}^{\mathbb{A}^1}(k)} &\simeq [\Sigma^n Z_+ \wedge \mathbb{G}^{\wedge n}, \mathbf{B}^{nis} G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \simeq [\Sigma^{n-1} Z_+ \wedge \mathbb{G}^{\wedge n}, G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \\ &\simeq [\Sigma^{n-1} Z_+ \wedge \mathbb{G}^{\wedge n-1}, \Omega_{\mathbb{G}} G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \simeq pt \text{ (since } \Omega_{\mathbb{G}} G = G_{-1} = 1). \end{aligned}$$

□

Remark 3.3.10. If one has an independent proof of Corollary 4.6 in [BHS23], then, together with Lemma 3.3.9 here and Corollary 3.8 from *loc.cit.*, one may deduce Theorem 4.5 in *loc.cit.*

Motivic null spaces

We now recall the notion of motivic null spaces from [ABH23]. A motivic space is said to be $S^{p,q}$ -null if it is local with respect to the motivic sphere $S^{p,q}$. In the special case of $S^{2,1}$ (and $S^{1,1}$), the definition can be restated as follows:

Definition 3.3.11. A space $\mathcal{X} \in \mathcal{P}(S)$ is said to be motivically $S^{2,1}$ (resp. $S^{1,1}$) null if it is $\mathbb{A}^1$ -local, Nisnevich-local, and $\mathbb{P}^1$ -local (resp. $\mathbb{G}_m$ -local) (see Theorem 2.2.1). We denote the corresponding localization functors by $L^{2,1} : \mathcal{P}(S) \rightarrow L^{2,1}\mathcal{P}(S)$ (and $L^{1,1} : \mathcal{P}(S) \rightarrow L^{1,1}\mathcal{P}(S)$, respectively).

We now discuss the main result leading to the announced model of L_{bir} :

Theorem 3.3.12. Let k be a perfect field. The following conditions on a Nisnevich-locally connected pointed motivic space $\mathcal{X}$ over k are equivalent:

1. $\mathcal{X}$ is birational local.
2. $\tau_{\leq i}^{nis} \mathcal{X}$ is birational local for all $i \geq 0$.
3. $\pi_i^{nis} \mathcal{X}$ is birational for all $i \geq 0$.
4. $(\pi_i^{nis} \mathcal{X})_{-1} = 0$ for all $i \geq 0$.
5. $\mathcal{X}$ is $\mathbb{G}_m$ local (equivalently, $S^{1,1}$ null).
6. $\mathcal{X}$ is $\mathbb{P}^1$ local (equivalently, $S^{2,1}$ null).

Proof. The equivalence of the first three conditions is Proposition 3.3.5. $3 \iff 4$ follows from Lemma 3.3.9 since $\pi_i^{nis} \mathcal{X}$ are strongly $\mathbb{A}^1$ -invariant [Mor12, Corollary 6.2]. $6 \iff 4 \iff 5$ follows from [ABH23, Corollary 3.1.21]. Indeed, for a motivic space, $\mathbb{G}_m$ -locality is equivalent to $S^{1,1}$ -nullity, which, by the above-mentioned corollary [ABH23, Corollary 3.1.21], is equivalent to the vanishing $(\pi_i^{nis} \mathcal{X})_{-1} = 0$ for $i \geq 0$ (the case $i = 0$ is trivially true due to the connectivity of $\mathcal{X}$). Similarly, $\mathbb{P}^1$ -locality is equivalent to $S^{2,1}$ -nullity, which, by the conditions in *loc.cit.*, is equivalent to the vanishing $(\pi_i^{nis} \mathcal{X})_{-1} = 0$ for $i \geq 1$ (again, the case $i = 0$ is trivially true by the connectivity of $\mathcal{X}$). □

Remark 3.3.13. Note that the results referred to in the proof, namely [ABH23, Corollary 3.1.21] do not use any explicit model for nullification and follow immediately from Morel's result: $\Omega_{\mathbb{G}} \pi_i^{\mathbb{A}^1} \simeq \pi_i^{\mathbb{A}^1} \Omega_{\mathbb{G}}$ [Mor12, Theorem 6.13].

Remark 3.3.14. Since $\mathcal{H}^b(k) \subset \mathcal{H}^{rat}(k) \subset \mathcal{H}^{\mathbb{A}^1}(k) \cap \mathcal{H}^{\mathbb{P}^1}(k)$ [see Corollary 2.2.9], when k is a perfect field, the above theorem can be restated as saying that the inclusions restrict to equalities

$$\tau_{\geq 1}^{nis} \mathcal{H}^b(k) = \tau_{\geq 1}^{nis} \mathcal{H}^{rat}(k) = \tau_{\geq 1}^{nis} (\mathcal{H}^{\mathbb{A}^1}(k) \cap \mathcal{H}^{\mathbb{P}^1}(k)).$$

The following counterexample shows that these equalities need not hold without the connectivity assumption.

Counterexample 3.3.15. By [Mai26a, Example 2.2.18], the stack of groupoids of finite etale schemes, denoted $\text{Fét} := \text{Fét}^{\simeq}$, is Nisnevich-local and $\mathbb{P}^1$ -local. When the base is the field of complex numbers, we claim that this is $\mathbb{A}^1$ -local as well. To see this, it suffices to show that for any smooth scheme $X/\mathbb{C}$, the canonical map $\text{Fét}(X) \rightarrow \text{Fét}(X \times_{\mathbb{C}} \mathbb{A}_{\mathbb{C}}^1)$ induced by the projection $X \times_{\mathbb{C}} \mathbb{A}_{\mathbb{C}}^1 \rightarrow X$ is an equivalence. This follows from the Riemann existence theorem [Mil80, Theorem 3.4] and the $\mathbb{C}$ -invariance of finite coverings of topological spaces. However, Fét is not birational local. For example, consider the dense open immersion $\mathbb{G}_{m,\mathbb{C}} \hookrightarrow \mathbb{A}_{\mathbb{C}}^1$. By [Sta26, Lemma 0BQF], we know that $\text{Fét}_{\mathbb{A}_{\mathbb{C}}^1} \rightarrow \text{Fét}_{\mathbb{G}_{m,\mathbb{C}}}$ is fully faithful. Hence, if $\text{Fét}(\mathbb{A}_{\mathbb{C}}^1) \rightarrow \text{Fét}(\mathbb{G}_{m,\mathbb{C}})$ were an equivalence of groupoids, then $\text{Fét}_{\mathbb{A}_{\mathbb{C}}^1} \rightarrow \text{Fét}_{\mathbb{G}_{m,\mathbb{C}}}$ would be so as well. But by the Riemann Existence Theorem again, this is impossible, since $\pi_1^{top}(\mathbb{G}_{m,\mathbb{C}}(\mathbb{C})) = \mathbb{Z}$ while $\pi_1^{top}(\mathbb{A}_{\mathbb{C}}^1(\mathbb{C})) = 0$. The problem is that Fét is not Nisnevich locally connected. Indeed, $\text{Fét}(\mathbb{C}) = \text{Fét}(\mathbb{C})^{\simeq} = \text{FinSet}^{\simeq}$, so $\pi_0^{nis} \text{Fét}$ evaluated at the base point $\text{Spec} \mathbb{C}$ equals to $\pi_0(\text{FinSet}^{\simeq}) = \mathbb{Z} \neq *$.

3.4 On finding a model for L_{bir}

In this section, we will finally construct a model for L_{bir} on motivically connected spaces. Before that, we take a detour to establish the required connectivity result for null spaces. The authors of [ABH23] deduce a connectivity result for $L^{p,q}$ in general; however, their formula for $L^{p,q}$, denoted $(-)^{p,q}_{\infty}$, is obtained only via the small object argument. We emphasize that in the special case of $S^{2,1}$ (and $S^{1,1}$) localization, it is possible to construct an explicit model (over an arbitrary base), and, in fact, one may immediately deduce a connectivity result from it. In an upcoming work, we will study the $\mathbb{P}^1$ motivic homotopy theory in greater detail [PHT Mai26a]. We recall the relevant results from there and note that they apply equally to $\mathbb{G}_m$ localization and, for that matter, to localization at any smooth scheme with a rational point. The reader may use the connectivity result in [ABH23, Corollary 3.1.26]. Our result is stronger in that it holds over arbitrary base schemes and yields an explicit surjection of path components.

Let us recall the following results from [Mai26a]:

Theorem 3.4.1 (Mai26a, Theorem 2.1.1, Lemma 2.1.4). *Let Y/S be a smooth scheme and $\mathcal{X} \in \mathcal{P}(S)$.*

1. *Let $\delta_{\mathbb{P}^1}^{\bullet} \times Y$ be the sub-diagram of the slice category $(\text{Sm}_S)_{/Y}$ defined by $\delta_{\mathbb{P}^1}^n \times Y := (\mathbb{P}_S^1)^{\times n} \times_S Y$, with structure maps given by projection to Y . The $\mathbb{P}^1$ -localization functor on $\mathcal{P}(S)$ is given by the following colimit formulae:*

$$X \mapsto L_{\mathbb{P}^1} \mathcal{X}(Y) = \text{colim}_{(\delta_{\mathbb{P}^1}^{\bullet} \times Y)^{op}} \mathcal{X}(\delta_{\mathbb{P}^1}^n \times Y)$$

2. *The canonical map $\pi_0 \mathcal{X} \rightarrow \pi_0 L_{\mathbb{P}^1} \mathcal{X}$ is a surjective morphism of presheaves of sets on Sm_S .*

This leads to the following explicit model of $L^{2,1}$:

Theorem 3.4.2. *A model for $S^{2,1}$ motivic localization is given by :*

$$L^{2,1} := L_{nis}^{\mathbb{A}^1, \mathbb{P}^1} \simeq \operatorname{colim}_{n \rightarrow \infty} (L_{nis} \circ L_{\mathbb{A}^1} \circ L_{\mathbb{P}^1})^n \simeq \operatorname{colim}_{n \rightarrow \infty} (L_{nis}^{\mathbb{A}^1} \circ L_{nis}^{\mathbb{P}^1})^n \simeq \operatorname{colim}_{n \rightarrow \infty} (L_{nis}^{\mathbb{P}^1} \circ L_{nis}^{\mathbb{A}^1})^n$$

where $L_{\mathbb{P}^1} \mathcal{X}(Y) = \operatorname{colim}_{(\delta_{\mathbb{P}^1}^\bullet \times Y)^{op}} \mathcal{X}(\delta_{\mathbb{P}^1}^m \times_S Y)$ and $L_{\mathbb{A}^1} \mathcal{Y}(Y) = |\mathcal{Y}(\Delta_{\mathbb{A}^1}^\bullet \times Y)|$.

Proof. The method is similar to Lemma 3.1.3. The space $\operatorname{colim}_{n \rightarrow \infty} (L_{nis}^{\mathbb{A}^1} \circ L_{nis}^{\mathbb{P}^1})^n \mathcal{X}$, associated with an S -space $\mathcal{X}$, is the colimit of the even-numbered subdiagram of the following diagram:

$$L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow L_{nis}^{\mathbb{P}^1} L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow \dots$$

It follows that the colimit of the whole diagram is both $\mathbb{A}^1$ -local and Nisnevich-local. Similarly, by considering the odd-numbered (cofinal) subdiagram, it follows that the colimit is $\mathbb{P}^1$ -local as well. The same arguments apply verbatim to the other colimit $\operatorname{colim}_{n \rightarrow \infty} (L_{nis}^{\mathbb{P}^1} \circ L_{nis}^{\mathbb{A}^1})^n \mathcal{X}$.

For the first colimit, one considers the $3\mathbb{N}$, $3\mathbb{N} + 1$, and $3\mathbb{N} + 2$ subdiagrams of the diagram: $L_{\mathbb{P}^1} \mathcal{X} \rightarrow L_{\mathbb{A}^1} L_{\mathbb{P}^1} \mathcal{X} \rightarrow L_{nis} L_{\mathbb{A}^1} L_{\mathbb{P}^1} \mathcal{X} \rightarrow L_{\mathbb{P}^1} L_{nis} L_{\mathbb{A}^1} L_{\mathbb{P}^1} \mathcal{X} \rightarrow L_{\mathbb{A}^1} L_{\mathbb{P}^1} L_{nis} L_{\mathbb{A}^1} L_{\mathbb{P}^1} \mathcal{X} \rightarrow \dots$ and proceeds similarly. $\square$

Corollary 3.4.3 (see ABH23, Corollary 3.1.26 for a weaker version). *Let S be Qcqs and $\mathcal{X} \in \mathcal{P}(S)$. Then there is a canonical surjection $\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L^{2,1} \mathcal{X}$. This implies that $L^{2,1}$ preserves Nisnevich connectivity.*

Proof. From Theorem 3.4.1 (2), it follows that $\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L_{\mathbb{P}^1} \mathcal{X}$ is surjective [see Mai26a, Proposition 2.1.5 for details]. By [MV99, Corollary 2.3.22], $\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L_{\mathbb{A}^1} \mathcal{X}$ is also surjective.

The claim then follows immediately from the formulas for $L^{2,1}$ given in the previous theorem, Theorem 3.4.2. For example, $L^{2,1} \mathcal{X}$ is given by the colimit of the following diagram:

$$\mathcal{X} \rightarrow L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow L_{nis}^{\mathbb{P}^1} L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow \dots$$

Since π_0^{nis} preserves colimits, we see that $\pi_0^{nis} L^{2,1} \mathcal{X}$ is the colimit of the following diagram

$$\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow \pi_0^{nis} L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow \pi_0^{nis} L_{nis}^{\mathbb{P}^1} L_{nis}^{\mathbb{A}^1} L_{nis}^{\mathbb{P}^1} \mathcal{X} \rightarrow \dots$$

Now, inductively applying the observations from the previous paragraph, we know that each map in the last diagram is an epimorphism of Nisnevich sheaves of sets. We conclude that the colimit $\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L^{2,1} \mathcal{X}$ is also an epimorphism. $\square$

Theorem 3.4.4. *Over a perfect field k , the restriction of $L_{bir} : \mathcal{P}(k)_\bullet \rightarrow \mathcal{H}^b(k)$ to $L^{2,1}$ -connected objects (resp. $L^{1,1}$ -connected objects) is equivalent to the restriction of $L^{2,1}$ (resp. $L^{1,1}$) to those objects.*

Proof. Suppose $\mathcal{X}$ is a space with a chosen base point such that $L^{2,1} \mathcal{X}$ (or $L^{1,1} \mathcal{X}$) is Nisnevich connected. By Theorem 3.3.12, $L^{2,1} \mathcal{X}$ (or $L^{1,1} \mathcal{X}$) is birational local. To conclude, it suffices to note that $L^{2,1}$ (or $L^{1,1}$) equivalences are birational equivalences. Indeed, then

the localization map $\mathcal{X} \rightarrow L^{2,1}\mathcal{X}$ (or $\mathcal{X} \rightarrow L^{1,1}\mathcal{X}$) is a birational equivalence to a birational local space, and hence is a model of birational localization. But since $\mathbb{A}^1, \mathbb{P}^1, \mathbb{G}_m$ are rational it is obvious that $L^{2,1}$ and $L^{1,1}$ equivalences are birational equivalences [Corollary 2.2.9 (1)]. $\square$

Corollary 3.4.5. *In the previous theorem, it suffices to assume that $\mathcal{X}$ is $\mathbb{A}^1$ -connected, $\mathbb{P}^1$ -connected, or even Nisnevich-connected.*

Proof. By Corollary 3.4.3, Nisnevich connectivity implies $L^{2,1}$ -connectivity. When $\mathcal{X}$ is $\mathbb{A}^1$ -connected, replace it with the Nisnevich-connected space $L_{mot}\mathcal{X}$. Similarly, when $\mathcal{X}$ is $\mathbb{P}^1$ -connected, replace it with the Nisnevich-connected space $L_{pmot}\mathcal{X}$. $\square$

Corollary 3.4.6. *On pointed $\mathbb{A}^1$ -connected spaces over perfect fields, the birational localization functor coincides with $L^{2,1}$ and $L^{1,1}$. Thus,*

$$L_{bir}\Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}(k)} \simeq \operatorname{colim}_{n \rightarrow \infty} (L_{nis} \circ L_{\mathbb{A}^1} \circ L_{\mathbb{P}^1})^n \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}} \simeq \operatorname{colim}_{n \rightarrow \infty} (L_{nis} \circ L_{\mathbb{A}^1} \circ L_{\mathbb{G}_m})^n \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}}$$

where $L_{\mathbb{P}^1}\mathcal{X}(Y) = \operatorname{colim}_{(\delta_{\mathbb{P}^1}^n \times Y)^{op}} \mathcal{X}(\delta_{\mathbb{P}^1}^n \times Y)$ and $L_{\mathbb{A}^1}\mathcal{Y}(Y) = |\mathcal{Y}(\Delta_{\mathbb{A}^1}^\bullet \times Y)|$. It follows that L_{bir} takes $\mathbb{A}^1$ connected spaces to Nisnevich connected spaces.

Corollary 3.4.7. *Over a perfect field, the birational localization functor preserves Nisnevich local connectivity.*

Proof. If $\mathcal{X}$ is Nisnevich connected, by Corollary 3.4.5, a choice of a base point of $\mathcal{X}$ (see Remark 3.5.6) yields $L_{bir}\mathcal{X} \simeq L^{2,1}\mathcal{X}$. But Corollary 3.4.3 implies that $L^{2,1}\mathcal{X}$ is Nisnevich connected. Thus $L_{bir}\mathcal{X}$ is Nisnevich connected as well. $\square$

Remark 3.4.8. From Construction 3.1.17 of [ABH23], we do have a formula for $S^{2,1}$ or $S^{1,1}$ localization, and hence,

$$L_{bir}\Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}} \simeq [(-)_{\infty}^{1,1}] \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}} \simeq [(-)_{\infty}^{2,1}] \Big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}}$$

Note, however, that these formulas are not explicit and are obtained using a small-object argument.

3.5 Connectivity of L_{bir}

In Corollary 3.4.7 of the last section, we saw that over a perfect field, L_{bir} preserves Nisnevich connectivity. Let us now compare this to the case of L_{mot} . Recall that, over an arbitrary Qcqs base, the singular model of motivic localization yields a canonical surjection $\pi_0^{nis} \rightarrow \pi_0^{\mathbb{A}^1}$, which implies a connectivity property for L_{mot} over an arbitrary Qcqs base scheme. For birational localization, we constructed a model in the previous section. However, it has two primary drawbacks: first, it works only for connected spaces, and second, it works only over perfect fields. Unfortunately, it seems less plausible to have such a mechanism for all spaces or over all bases. In this section, we will take an entirely different route to prove, in fact, a stronger version of connectivity for the birational localization functor. The key will be the commutation of L_{bir} with the bar construction, as obtained in §3.2. We continue to assume that S is Qcqs. We start with the primary observation,

Corollary 3.5.1. *A birational local space is birationally connected if and only if it is $\mathbb{A}^1$ -connected if and only if it is Nisnevich-connected if and only if it is sectionwise connected.*

Proof. Clearly, if $\mathcal{X}$ is birational local, then $\pi_0^b \mathcal{X} \cong \pi_0 \mathcal{X} \cong \pi_0^{nis} \mathcal{X}$ (see Corollary 2.1.9 for the last isomorphism). By Corollary 2.2.11, it follows that $\mathcal{X} \simeq L_{mot} \mathcal{X}$. Applying π_0^{nis} to this equivalence yields $\pi_0^{nis} \mathcal{X} \cong \pi_0^{\mathbb{A}^1} \mathcal{X}$. Thus $\pi_0^b \mathcal{X} \cong \pi_0 \mathcal{X} \cong \pi_0^{nis} \mathcal{X} \cong \pi_0^{\mathbb{A}^1} \mathcal{X}$. $\square$

Thus, we shall say that a space $\mathcal{X}$ is birationally connected if $L_{bir} \mathcal{X}$ is (sectionwise/Nisnevich locally/ $\mathbb{A}^1$)-connected.

Corollary 3.5.2. *If σ is a topology coarser than the Nisnevich topology over an arbitrary Qcqs base S , then L_{bir} takes σ locally $n(\geq 0)$ -connected pointed S -spaces to sectionwise n -connected S -spaces.*

Proof. From the loop-path fiber sequence, the existence of a base point for $\mathcal{X} \in \tau_{\geq 1} \mathcal{P}(S)$ yields an equivalence $\mathcal{X} \simeq B_\sigma \Omega(\mathcal{X})$. By induction, if $\mathcal{X}$ is σ -locally n -connected, then $\mathcal{X} \simeq B_\sigma^{n+1} \Omega^{n+1} \mathcal{X}$. But then $L_{bir} \mathcal{X} \simeq L_{bir} B_\sigma^{n+1} \Omega(\mathcal{X}) \simeq B^{n+1} L_{bir} \Omega(\mathcal{X})$ [by Lemma 3.2.3]. By construction, B^{n+1} takes values in (sectionwise) n -connected objects and we are done. $\square$

Corollary 3.5.3. *L_{bir} takes pointed $\mathbb{A}^1$ - n -connected spaces to (sectionwise) n -connected spaces. In particular, pointed $\mathbb{A}^1$ -connected spaces are birationally connected.*

Proof. Assume $\mathcal{X}$ is pointed $\mathbb{A}^1$ - n -connected, i.e., $\mathcal{Y} := L_{mot} \mathcal{X}$ is a Nisnevich n -connected pointed space. By the previous corollary applied to the Nisnevich topology, $L_{bir} \mathcal{Y}$ is sectionwise n -connected. Since motivic equivalences are birational equivalences [see Corollary 2.2.11], we have $L_{bir} \mathcal{X} \simeq L_{bir} L_{mot} \mathcal{X} = L_{bir} \mathcal{Y}$. It thus follows that $L_{bir} \mathcal{X}$ is sectionwise n -connected. $\square$

Remark 3.5.4 ((-1, -2)-connectivity). Since every object is (-2)-connected, like any localization, L_{bir} preserves (-2)-connectivity. Similarly, a (-1)-connectivity statement is also true for any localization. For example, if $U \rightarrow \mathcal{X}$ is a non trivial section of a space $\mathcal{X}$ over some nonempty scheme U , then $U \rightarrow \mathcal{X} \rightarrow L_{bir} \mathcal{X}$ provides a (non-empty) section of $L_{bir} \mathcal{X}$.

Remark 3.5.5. One reason such a method of proving connectivity, as in Corollary 3.5.2, is not feasible in the motivic case is that the 2nd statement in Lemma 3.2.3 fails in the $\mathbb{A}^1$ -motivic case, see Remark 3.2.5. This is why, in the $\mathbb{A}^1$ -motivic case, an explicit formula for the motivic localization functor, which ensures surjectivity on π_0^{nis} , was needed to obtain an unstable connectivity result [MV99, Corollary 3.22]. Though we must mention that the surjectivity at π_0^{nis} ensures that $\eta : 1 \rightarrow L_{mot}$ is an objectwise effective epimorphism, which is an important tool for importing several results from the Nisnevich topos to the motivic homotopy category (see, for example, [Mai26b, Corollary 3.4.5]). However, for birational localization, it is unclear whether the canonical map $\pi_0^{nis} \mathcal{X} \rightarrow \pi_0^{nis} L_{bir} \mathcal{X} = \pi_0 L_{bir} \mathcal{X}$ is surjective, unlike its $\mathbb{A}^1$ -motivic counterpart. As a result, the birational localization yields weaker results (see [Mai26b, Corollary 4.4.6]).

Remark 3.5.6. When the topology σ is such that the base scheme S admits no nontrivial σ -cover, we can remove the pointed condition from each of the statements above. This is

always the case for the discrete topology, or when S is a point for the topology σ . For example, this holds when $\sigma = zar$ and S is the spectrum of a local ring, or when $\sigma = nis$ and S is the spectrum of a henselian local ring. Indeed, in these cases $\pi_0^\sigma \mathcal{X}(S) = \pi_0 \mathcal{X}(S)$. However, this is not always the case. For example, this might fail when $S = \mathbb{P}_k^1$ and $\sigma = zar$. An example of a topology finer than the Nisnevich topology where this fails is the étale topology over the spectrum of a non-separably closed field.

Corollary 3.5.7. *Let S be the spectrum of a henselian local ring. Then,*

1. L_{bir} takes Nisnevich locally n -connected S -spaces to sectionwise n -connected S -spaces.
2. L_{bir} takes $\mathbb{A}^1$ - n -connected S -spaces to sectionwise n -connected S -spaces.
3. $\mathbb{A}^1$ -connected spaces over S are birationally connected.

Proof. Because S is henselian, for any $\mathcal{X} \in \mathcal{P}(S)$ we have $\pi_0 \mathcal{X}(S) \simeq \pi_0^{nis} \mathcal{X}(S)$. Hence, if $\mathcal{X}$ is Nisnevich connected, then $\pi_0 \mathcal{X}(S) \simeq \pi_0^{nis} \mathcal{X}(S) = *$. We may thus choose a section $S \rightarrow \mathcal{X}$ to make $\mathcal{X}$ pointed. So (1) follows from Corollary 3.5.2. For (2), use $\mathcal{Y} := L_{mot} \mathcal{X}$ as in the proof of Corollary 3.5.2. (3) is a special case of (2). $\square$

3.6 Birational homotopy groups of spaces

We will use the notation $\pi_i^b := \pi_i L_{bir}$ and call these the birational homotopy group presheaves. By Proposition 2.1.8, these presheaves on Sm_S (see Definition Definition 2.1.6) are therefore automatically Nisnevich sheaves by Theorem 2.1.3. Consequently, we have $\pi_i^b = \pi_i^{nis} L_{bir}$, aligning with the conventions in motivic homotopy theory. Throughout this section, we assume that σ is a topology coarser than the Nisnevich topology.

Birational connective covers and universal properties of π_i^b :

Let $\mathcal{X} \in \mathcal{P}_\sigma(S)_\bullet$ and $\tau_{>n}^\sigma \mathcal{X} \rightarrow \mathcal{X} \rightarrow \tau_{\leq n}^\sigma \mathcal{X}$ be the n -th slice of the Postnikov tower of $\mathcal{X}$.

Lemma 3.6.1. *If $\mathcal{X}$ is a birational local space, then $\tau_{>n}^\sigma \mathcal{X}$ is birational local.*

Proof. When $\mathcal{X}$ is birational local, it follows that so is $\tau_{>1} \mathcal{X}$ (use, for example, Proposition 2.1.8). But then, for any topology σ coarser than the Nisnevich topology, $\tau_{>1} \mathcal{X}$ is σ local as well [Theorem 2.1.3]. Therefore, $\tau_{>n}^\sigma \mathcal{X} := L_\sigma \tau_{>n} \mathcal{X} \simeq \tau_{>1} \mathcal{X}$, and this last term is already birational local. $\square$

Corollary 3.6.2. *The (sectionwise) simply connected covering space of a birational local space is birational local. We may call this covering space a ‘birationally simply connected cover’.*

Proof. This is the $n = 1$ case of the lemma above. $\square$

Since $\tau_{>n}^\sigma L_{bir}$ is birational local by Lemma 3.6.1, the natural map $\tau_{>n}^\sigma \rightarrow \tau_{>n}^\sigma L_{bir}$ descends to a map $L_{bir} \tau_{>n}^\sigma \rightarrow \tau_{>n}^\sigma L_{bir}$. Similarly, by Proposition 3.3.5 there is a canonical map $L_{bir} \tau_{\leq n}^\sigma \rightarrow \tau_{\leq n}^\sigma L_{bir}$.

Proposition 3.6.3. *Let $\mathcal{X}$ be a pointed σ -locally 0-connected S -space such that for some $n \geq 0$ the S -space $\tau_{\leq n}^\sigma \mathcal{X}$ is birational (this happens precisely when $\pi_i^\sigma \mathcal{X}$ are birational*

local for all $i \leq n$, see Corollary 2.1.9). Then, for all $i \leq n$, the canonical morphisms $L_{bir}\tau_{\leq i}^\sigma \rightarrow \tau_{\leq i}^\sigma L_{bir}$ and $L_{bir}\tau_{> i}^\sigma \rightarrow \tau_{> i}^\sigma L_{bir}$ are equivalences.

Proof. Since $\tau_{\leq i}^\sigma \mathcal{X}$ is σ -locally connected, we obtain a fiber sequence

$$L_{bir}\tau_{> i}^\sigma \mathcal{X} \rightarrow L_{bir}\mathcal{X} \rightarrow L_{bir}\tau_{\leq i}^\sigma \mathcal{X}$$

by applying Lemma 3.2.11. But $L_{bir}\tau_{\leq i}^\sigma \mathcal{X} \simeq \tau_{\leq i}^\sigma \mathcal{X}$ for $i \leq n$ because $\tau_{\leq n}^\sigma \mathcal{X}$ is birational. Because $\tau_{\leq i}^\sigma \mathcal{X}$ is connected and $\tau_{> i}^\sigma \mathcal{X}$ is i -connected (in $\mathcal{P}_\sigma(S)$), by higher connectivity [Corollary 3.5.2], $L_{bir}\tau_{\leq i}^\sigma \mathcal{X}$ is connected and $L_{bir}\tau_{> i}^\sigma \mathcal{X}$ is i -connected. From the long exact sequence of homotopy groups of the fiber sequence $L_{bir}\tau_{> i}^\sigma \mathcal{X} \rightarrow L_{bir}\mathcal{X} \rightarrow L_{bir}\tau_{\leq i}^\sigma \mathcal{X}$, this yields $L_{bir}\tau_{\leq i}^\sigma \mathcal{X} \simeq \tau_{\leq i}^\sigma L_{bir}\mathcal{X}$.

On the other hand, we compare the fiber sequence in the first line with the fiber sequence

$$\tau_{> i}^\sigma L_{bir}\mathcal{X} \rightarrow L_{bir}\mathcal{X} \rightarrow \tau_{\leq i}^\sigma L_{bir}\mathcal{X}$$

given by the n -th slice of the Postnikov tower of $L_{bir}\mathcal{X}$.

$$\begin{array}{ccccc} L_{bir}\tau_{> i}^\sigma \mathcal{X} & \longrightarrow & L_{bir}\mathcal{X} & \longrightarrow & L_{bir}\tau_{\leq i}^\sigma \mathcal{X} \\ \downarrow & & \downarrow & & \downarrow \\ \tau_{> i}^\sigma L_{bir}\mathcal{X} & \longrightarrow & L_{bir}\mathcal{X} & \longrightarrow & \tau_{\leq i}^\sigma L_{bir}\mathcal{X} \end{array}$$

Since both the middle and right vertical morphisms are equivalences, we deduce that the left vertical map is also an equivalence. $\square$

Corollary 3.6.4. *Let $\mathcal{X} \in \mathcal{P}_\sigma(S)_{\geq 1}$ be pointed, with a σ -locally simply connected cover $\tilde{\mathcal{X}}$. If $\pi_1^\sigma \mathcal{X}$ is birational, then $L_{bir}\tilde{\mathcal{X}} \rightarrow L_{bir}\mathcal{X}$ is a birationally simply connected covering. It follows that there is an isomorphism $\pi_1 \mathcal{X} \simeq \pi_1^b \mathcal{X} \simeq \text{Aut}_{L_{bir}\mathcal{X}}(L_{bir}\tilde{\mathcal{X}})$.*

Proof. Applying Proposition 3.6.3 with $n = 1$ it follows that $L_{bir}\tau_{> 1}^{nis} \mathcal{X} \rightarrow \tau_{> 1}^{nis} L_{bir}\mathcal{X}$ is an equivalence. The claim follows from the fact that for a pointed σ -locally connected space $\mathcal{Y}$, $\tau_{> 1}^\sigma \mathcal{Y}$ is its unique pointed simply connected cover. $\square$

Remark 3.6.5. In principle, similar to Morel's result in his book [Mor12, Theorem 7.8], one would like a statement about the existence of birational covering spaces over birationally connected spaces, and that too without the birationality assumption of π_1 . Unfortunately, because we do not know whether L_{bir} is locally cartesian, we are unable to establish this in such generality. However, Corollary 3.2.14 suggests that this is possible over $\mathbb{A}^1$ -connected objects. This will require categorical discussions of covering spaces, which we will develop in an upcoming work [Mai26b, §2.4]. As a consequence, we will be able to construct the desired birational result over $\mathbb{A}^1$ -connected spaces. In the same paper, we will also reprove some versions of Morel's results on $\mathbb{A}^1$ covering spaces, adapted to the ∞ -categorical language.

Theorem 3.6.6. *Let $\mathcal{X}$ be a pointed σ -locally 0-connected S -space such that, for some $n \geq 0$, the space $\tau_{\leq n}^\sigma \mathcal{X}$ is birational (which occurs precisely when $\pi_i^\sigma \mathcal{X}$ are birational local for all $i \leq n$; see Corollary 2.1.9). Then the morphism $\pi_{n+1}^\sigma \mathcal{X} \rightarrow \pi_{n+1}^b \mathcal{E}$ is universal among morphisms to a birational sheaf of groups when $n = 0$ and to a birational sheaf of abelian groups when $n \geq 1$.*

Proof. Let B be a birational sheaf of (abelian if $n \geq 1$) groups. Since $\tau_{>n}^\sigma \mathcal{X}$ is σ -locally n -connected by definition, it admits an $(n+1)$ -fold delooping. In other words, we have an equivalence $\tau_{>n}^\sigma \mathcal{X} \simeq B_\sigma^{n+1} \Omega^{n+1} \tau_{>n}^\sigma \mathcal{X}$. But then

$$\mathrm{Map}_{\mathcal{P}_\sigma(S)}(\tau_{>n}^\sigma \mathcal{X}, \mathbf{B}_\sigma^{n+1} B) \simeq \mathcal{E}_{n+1}\text{-Mon}_{\mathcal{P}_\sigma(S_{m_S})}(\Omega^{n+1} \tau_{>n}^\sigma \mathcal{X}, B) \quad (3)$$

$$\cong (n+1)\text{-Grp}_S^\sigma(\pi_0 \Omega^{n+1} \tau_{>n}^\sigma \mathcal{X}, B) \text{ [since } B \text{ is discrete]} \quad (4)$$

$$\cong (n+1)\text{-Grp}_S^\sigma(\pi_{n+1}^\sigma \mathcal{X}, B) \quad (5)$$

On the other hand, since the space $B_\sigma^{n+1} B = K(B, n+1)$ is birational local [see Lemma 3.2.6], the starting infinity groupoid is identical to $\mathrm{Map}_{\mathcal{H}^b(S)}(L_{\mathrm{bir}} \tau_{>n}^\sigma \mathcal{X}, \mathbf{B}_\sigma^{n+1} B)$. By Proposition 3.6.3, this mapping space can be further identified with $\mathrm{Map}_{\mathcal{P}_\sigma(S)}(\tau_{>n}^\sigma L_{\mathrm{bir}} \mathcal{X}, \mathbf{B}_\sigma^{n+1} B)$. By the same line of identifications as in the above computation (3-5), this last term is equivalent to $(n+1)\text{-Grp}_S^\sigma(\pi_{n+1}^\sigma L_{\mathrm{bir}} \mathcal{X}, B)$. Again, because $L_{\mathrm{bir}} \mathcal{X}$ is birational $\pi_{n+1} L_{\mathrm{bir}} \mathcal{X} \simeq \pi_{n+1}^\sigma L_{\mathrm{bir}} \mathcal{X}$. Thus, $(n+1)\text{-Grp}_S^\sigma(\pi_{n+1}^\sigma L_{\mathrm{bir}} \mathcal{X}, B) \cong (n+1)\text{-Grp}_S^b(\pi_{n+1}^b \mathcal{X}, B)$. One may now use the Yoneda lemma to conclude. (here, $(n+1)\text{-Grp}_S^\sigma$ is the category of σ -sheaves of $(n+1)$ -group objects; when $n = 0$, this is simply Grp_S^σ , whereas for $n \geq 1$, an Eckmann-Hilton argument shows that these are all equivalent to Ab_S^σ .) $\square$

The π_0 case does not require any condition:

Proposition 3.6.7. *For any $\mathcal{X} \in \mathcal{P}(S)$, the natural map $\pi_0^\sigma \mathcal{X} \rightarrow \pi_0^b \mathcal{X}$ is the universal morphism to a birational (pre)sheaf of sets. In particular, replacing $\mathcal{X}$ by $L_{\mathrm{mot}} \mathcal{X}$ yields that $\pi_0^{\mathbb{A}^1} \mathcal{X} \rightarrow \pi_0^b \mathcal{X}$ is a birationalization.*

Proof. Since a birational sheaf of sets B is discrete, we compute $\mathrm{Map}_{\mathcal{P}(S_{m_S})}(\mathcal{X}, B) \simeq PSh_S(\pi_0 \mathcal{X}, B)$. Again, since B is birational as a space, the left-hand side is

$$\mathrm{Map}_{\mathcal{H}^b(S)}(L_{\mathrm{bir}} \mathcal{X}, B) \simeq \mathrm{Map}_{\mathcal{P}(S)}(L_{\mathrm{bir}} \mathcal{X}, B) \simeq PSh_S(\pi_0 L_{\mathrm{bir}} \mathcal{X}, B) = Shv_S^b(\pi_0^b \mathcal{X}, B).$$

These equivalences induce a canonical identification $PSh_S(\pi_0 \mathcal{X}, B) \cong Shv_S^b(\pi_0^b \mathcal{X}, B)$. Formally, this implies that $\pi_0^b \mathcal{X}$ is the birationalization of $\pi_0 \mathcal{X}$. Since birational presheaves are σ -sheaves, this immediately guarantees (by the universal property of sheafification) that $\pi_0^b \mathcal{X}$ is the birationalization of $\pi_0^\sigma \mathcal{X}$ as well. For the last statement, use $\sigma = \mathrm{nis}$. $\square$

Remark 3.6.8. For a discrete presheaf of sets, this implies that $F \rightarrow \pi_0^b F$ provides a left adjoint to the inclusion $Shv_S^b \hookrightarrow Shv_S^\sigma$ (which has been studied over fields earlier in [Shi23], where it is denoted π_0^{br}). So the observations of the above proposition can be restated as

$$\pi_0^b \mathcal{X} \simeq \pi_0^b \pi_0 \mathcal{X} \simeq \pi_0^b \pi_0^{\mathrm{nis}} \mathcal{X} \simeq \pi_0^b \pi_0^{\mathbb{A}^1} \mathcal{X}.$$

Moreover, since both π_0 and L_{bir} preserve finite products [Theorem 3.2.2], we know that $\pi_0^b = \pi_0 L_{\mathrm{bir}}$ also preserves them. So in particular, π_0^b preserves group structures. Therefore, the conclusion of Theorem 3.6.6 can be restated as $\pi_{n+1}^b \mathcal{X} \cong \pi_0^b \pi_{n+1}^\sigma \mathcal{X}$.

Corollary 3.6.9. *Suppose $\mathcal{X}$ is a (motivically) pointed $\mathbb{A}^1$ - n -connected S -space. Then the canonical map $\pi_{n+1}^{\mathbb{A}^1} \mathcal{X} \rightarrow \pi_{n+1}^b \mathcal{X}$ is universal among maps to a birational sheaf of sets when $n = -1$; to a birational sheaf of groups when $n = 0$; and to a birational sheaf of abelian groups when $n \geq 1$; i.e., $\pi_{n+1}^b \mathcal{X} \simeq \pi_0^b(\pi_{n+1}^{\mathbb{A}^1} \mathcal{X})$.*

Proof. When $n = -1$, this is Proposition 3.6.7. For $n \geq 0$, apply Theorem 3.6.6 to the space $\mathcal{Y} := L_{mot}\mathcal{X}$ with the Nisnevich topology. By the given condition, $\tau_{\leq n}^{nis}\mathcal{Y} \simeq pt$ is trivially birational. It follows that $\pi_{n+1}^{nis}\mathcal{Y} \rightarrow \pi_{n+1}^b\mathcal{Y}$ is the universal birational sheaf. But this map is identical to the map in question, since $L_{bir}L_{mot} \simeq L_{bir}$. The equivalence $\pi_{n+1}^b\mathcal{X} \simeq \pi_0^b(\pi_{n+1}^{\mathbb{A}^1}\mathcal{X})$ (as groups) follows from Remark 3.6.8. $\square$

Remark 3.6.10. When the base is the spectrum of a perfect field k and $n \geq 0$, the above condition can be restated, using Lemma 3.3.9, as saying that the map $\pi_{n+1}^{\mathbb{A}^1}\mathcal{X} \rightarrow \pi_{n+1}^b\mathcal{X}$ is universal among maps to a Nisnevich sheaf of (abelian when $n \geq 1$) groups that is (strictly when $n \geq 1$) strongly- $\mathbb{A}^1$ -invariant and has trivial $\mathbb{G}_m$ -contraction.

Computing birational path component π_0^b :

Therefore, the transition from motivic homotopy (groups) to birational motivic homotopy (groups) is controlled by π_0^b of discrete S -spaces. We want to observe that in sufficiently nice scenarios, this behaves quite well. In fact, it turns out that computing the birational path component of any presheaf of spaces is easier than computing its $\mathbb{A}^1$ counterpart. To see this, recall that by Lemma 3.1.6, for $\mathcal{X} \in \mathcal{P}(S)$ we may write

$$L_{bir}\mathcal{X} \simeq \operatorname{colim}_{\substack{X \rightarrow \mathcal{X} \\ X \in Sm_S}} L_{bir}X'$$

where the colimit is computed in $\mathcal{P}(S)$ and is taken over the diagram of sections of $\mathcal{X}$ over X/S , and X' is a P -model up to a birational morphism, as listed in Lemma 3.1.6. Since $\pi_0 : \mathcal{P}(S) \rightarrow PSh_S$ preserves colimits, it follows that

$$\pi_0^b\mathcal{X} = \pi_0 L_{bir}\mathcal{X} \simeq \operatorname{colim}_{\substack{X \rightarrow \mathcal{X} \\ X/S \in Sm_S}} \pi_0^b X'$$

where the colimit is computed in the category of presheaves of sets on Sm_S . Thus, to compute $\pi_0^b\mathcal{X}$ for any $\mathcal{X}$, it suffices to know $\pi_0^b X'$ for X'/S with property P as listed in Lemma 3.1.6. For example, the referred theorem shows that P can always be taken to be the property that X/S is proper. Therefore, to compute π_0^b of any space, it suffices to know $\pi_0^b X$ for every proper scheme X/S . Moreover, when S is noetherian, condition 5 of Lemma 3.1.6 suggests that X'/S can also be taken to be projective. It turns out that, over fields, whenever X is proper, there is a very old classical model of $\pi_0^b X$, which we explain next.

For this, let us fix an arbitrary base field k . Given a proper scheme X/k , there is a birational sheaf, denoted $\pi_0^{b\mathbb{A}^1}X$, whose sections over a finitely generated separable field extension L/k is given by the set of (Manin's) R-equivalence classes of L -points of X [AM11, §6]. The method used to prove this existence is as follows. The authors first introduce the category $Shv_k^{b\mathbb{A}^1}$, whose objects are birational sheaves that are $\mathbb{A}^1$ -invariant. Then they prove that the restriction functor $Shv_k^{b\mathbb{A}^1} \rightarrow \mathcal{F}_k^r\text{-Sets}$, given by evaluation at the generic point, is fully faithful, with essential image consisting of sheaflike transcendental invariants (which they call $\mathbb{A}^1$ -invariant) data [AM11, Theorem 6.1.7]. Although it is not difficult to see that,

Lemma 3.6.11. *For a field k , the inclusion $Shv_k^{b\mathbb{A}^1} \subset Shv_k^b$ is an equality, and the functor $Snv_k^b \rightarrow \mathcal{F}_k^r\text{-Sets}$ given by evaluation at the generic point is fully faithful, with essential image consisting of sheaf-like transcendental invariant data (see [AM11, §6] for definition).*

Proof. This is because birational presheaves of sets are automatically $\mathbb{A}^1$ -invariant [Corollary 2.2.7]. $\square$

Finally, the authors of *loc.cit.* prove that, for a proper variety X/k , the assignment $L \mapsto X(L)/\sim$ of naive $\mathbb{A}^1$ -homotopy classes of L -points of X is a sheaflike transcendental invariant $\mathcal{F}_k^r$ -data [see Theorem 6.2.1 of *loc.cit.*]. We claim that this is exactly the candidate $\pi_0^b X$ we were looking for. This result was obtained for **smooth proper varieties** in [CR24, Theorem 3.4] using Kahn-Sujhata's localized category $S_b^{-1}Sm_k$. The proof below is new. It is formal and does not require $S_b^{-1}Sm_k$, and thus works for all proper schemes:

Theorem 3.6.12 (this is a generalization of [CR24, Theorem 3.4] and [CK23, Proposition 1.9.] to all proper schemes). *Let k be a field and X a proper scheme over k . Then there is a canonical isomorphism $\pi_0^b X \simeq \pi_0^{b\mathbb{A}^1} X$ of presheaves.*

Proof. By Proposition 3.6.7, we know that $X \cong \pi_0 X \rightarrow \pi_0^b X$ provides the birationalization of X . But by [AM11, Remark 6.2.2], for any proper scheme X , the canonical morphism $X \rightarrow \pi_0^{b\mathbb{A}^1} X$ is the universal object of $Shv_k^{b\mathbb{A}^1}$ associated to X . Because, $Shv_k^{b\mathbb{A}^1} = Shv_k^b$, $X \rightarrow \pi_0^{b\mathbb{A}^1} X$ is in fact, the universal birationalization. By comparing the universal properties, one immediately obtains the canonical isomorphism $\pi_0^b X \xrightarrow{\sim} \pi_0^{b\mathbb{A}^1} X$. $\square$

Corollary 3.6.13 (this is a generalization of [Koi22, Theorem 2.4]). *The assignment $\pi_0^{b\mathbb{A}^1}(-) : \text{Prop}_k \rightarrow Sh_k$ is a birational invariant, i.e., if $X \dashrightarrow Y$ is a birational morphism of proper schemes, then there is an isomorphism of sheaves $\pi_0^{b\mathbb{A}^1} X \simeq \pi_0^{b\mathbb{A}^1} Y$.*

Proof. Note that the assignment $\pi_0^b(-)_{|\text{Sch}_k} : \text{Sch}_k \rightarrow Sh_k$ given by $X \mapsto \pi_0^b(X) = \pi_0 L_{\text{bir}} X$, by construction, is a birational invariant. The claim then follows from the theorem above, Theorem 3.6.12. $\square$

Corollary 3.6.14. *The assignment $\mathbb{H}_0^{\mathbb{A}^1}(-) : \text{SmProp}_k \rightarrow Ab_k$ is a birational invariant.*

Proof. By [Koi22, Theorem 4.5], we know that if X is smooth and proper, then $\mathbb{H}_0^{\mathbb{A}^1}(X) \simeq \mathbb{Z}(\pi_0^{b\mathbb{A}^1} X)$. Thus, the claim follows from the above corollary. $\square$

Corollary 3.6.15. *For $X \in \text{Prop}_k$, the canonical map $\pi_0^{\mathbb{A}^1} X \rightarrow \pi_0^{b\mathbb{A}^1} X$ [AM11, Proposition 6.2.6] is initial among maps to a birational sheaf of sets.*

Proof. Combine Proposition 3.6.7 and Creffrom choudhury. $\square$

To summarize, for a k -Space $\mathcal{X}$ we have:

$$\pi_0^b \mathcal{X} = \pi_0 L_{\text{bir}} \mathcal{X} \simeq \text{colim}_{\substack{X \rightarrow X' \\ X \in Sm_k}} \pi_0^{b\mathbb{A}^1} X' \simeq \text{colim}_{\substack{X \hookrightarrow X \rightarrow \bar{X} \\ X \in Sm_k, \bar{X} \in Comp_k(X)}} \pi_0^{b\mathbb{A}^1} \bar{X} \quad (6)$$

where the colimit is taken over spans with $X \hookrightarrow \bar{X}$ a compactification of X . When S is noetherian, we might also consider projective models for X [see Lemma 3.1.6 (5)] and replace the colimit diagram in (6) with the diagram

$$\left(\begin{array}{c} \mathcal{X} \leftarrow X \hookrightarrow \bar{X} \leftarrow X'' \\ X'' \rightarrow X \text{ is a projective model of } \bar{X} \end{array} \right) \mapsto \pi_0^{b\mathbb{A}^1} X''$$

When k is moreover of characteristic 0, we may assume X'' is smooth and projective [see Lemma 3.1.6 (6)].

All of this applies equally to presheaves of sets. That is, for any presheaf F on Sm_k : $\pi_0^b F \simeq \operatorname{colim}_{F \leftarrow X \hookrightarrow \bar{X}} \pi_0^{b\mathbb{A}^1}(\bar{X})$. By Remark 3.6.8, we may thus replace the diagram in (6) by

$$\left(\begin{array}{c} \pi_0^{\mathbb{A}^1} \mathcal{X} \hookrightarrow X \rightarrow \bar{X} \\ X \in Sm_k, \bar{X} \in Comp_k(X) \end{array} \right) \mapsto \pi_0^{b\mathbb{A}^1}(\bar{X}).$$

Remark 3.6.16. Recall that the motivic homotopy category is also generated by sifted colimits of motivic models of smooth (affine) schemes. However, the tricks used in our birational case do not quite apply to the ordinary motivic case. The problem is multifold:

1. First, since the motivic homotopy category $\mathcal{H}^{\mathbb{A}^1}(S)$ is not closed under sifted colimits in $\mathcal{P}_{nis}(S)$ [Remark 3.1.2], the above-mentioned colimits are computed only in $\mathcal{H}^{\mathbb{A}^1}(S)$, not in $\mathcal{P}_{nis}(S)$. Therefore, π_0^{nis} need not preserve these colimits.
2. On the other hand, since $\pi_0^{\mathbb{A}^1}$ is not the internal 0-truncation of $\mathcal{H}^{\mathbb{A}^1}(S)$ [Ayo], applying even $\pi_0^{\mathbb{A}^1}$ might not help either.
3. Finally, even if one could manage this in a sufficiently nice setup, computing $\pi_0^{\mathbb{A}^1}$ is known to be extremely difficult, even for schemes. In fact, even over fields, there is no class of schemes known to date that generates the motivic homotopy category and for which $\pi_0^{\mathbb{A}^1}$ is completely known.

The observation of Theorem 3.6.12 makes π_0^b into a powerful device. So much so that it can detect connectivity of ‘proper’ motivic spaces (the last term means an object of the subcategory of $\mathcal{H}^{\mathbb{A}^1}(k)$ generated under filtered colimits by proper schemes).

Proposition 3.6.17. *Let $\mathcal{H}_{prop}^{\mathbb{A}^1}(k)$ be the full subcategory of $\mathcal{H}^{\mathbb{A}^1}(k)$ generated under filtered colimits by proper schemes. Then the restriction of the functor $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \rightarrow \mathcal{H}^b(k)$ to $\mathcal{H}_{prop}^{\mathbb{A}^1}(k)$ preserves and detects connectivity.*

Proof. Preservation of connectivity follows from Corollary 3.5.7(2).

To address detection, let $\mathcal{X}$ be a motivic local space given by the filtered colimit $\operatorname{colim}_{X \in \operatorname{Prop}/\mathcal{X}} L_{mot}(X)$ in $\mathcal{H}^{\mathbb{A}^1}(k)$. Since $\mathcal{H}^{\mathbb{A}^1}(k) \subset \mathcal{P}_{nis}(k)$ is closed under filtered colimits, we see that this colimit can be computed in $\mathcal{P}_{nis}(k)$. Since $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \rightarrow \mathcal{P}_{nis}(k)$ preserves filtered colimits (indeed, $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \rightarrow \mathcal{H}^b(k)$ is a left adjoint and $\mathcal{H}^b(k) \subset \mathcal{P}_{nis}(k)$ is closed under filtered colimits [see Theorem 3.1.1]), we deduce that the morphism $\mathcal{X} \rightarrow L_{bir}\mathcal{X}$ can be identified with the colimit

$$\operatorname{colim}_{X \in \operatorname{Prop}/\mathcal{X}} (L_{mot}X \rightarrow L_{bir}X)$$

in $\mathcal{P}_{nis}(k)$. Applying π_0^{nis} , we find that $\pi_0^{\mathbb{A}^1}\mathcal{X} \rightarrow \pi_0^b\mathcal{X}$ is the colimit

$$\operatorname{colim}_{X \in \operatorname{Prop}/\mathcal{X}} (\pi_0^{\mathbb{A}^1}X \rightarrow \pi_0^bX).$$

Since every X in the colimit above is proper, Theorem 3.6.12 implies that each term in the colimit is an isomorphism over $\mathcal{F}_k$ [Aso13, Proposition 6.2.6], and hence so is $\pi_0^{\mathbb{A}^1}\mathcal{X} \rightarrow \pi_0^b\mathcal{X}$. Thus, when $\pi_0^b\mathcal{X}$ is trivial, we have that $\pi_0^{\mathbb{A}^1}\mathcal{X}$ is trivial over $\mathcal{F}_k$. Since triviality of $\pi_0^{\mathbb{A}^1}\mathcal{X}$ can be detected over $\mathcal{F}_k$ [Bac24, Corollary 2.10.], we are done. $\square$

Corollary 3.6.18. *An (ind-)proper scheme over a field k is $\mathbb{A}^1$ -connected if and only if it is birationally connected.*

Remark 3.6.19 (this is a consequence of a private communication with Dr. Utsav Choudhary). The counter example of Theorem 2.2.10 is that of a three fold. For (proper) schemes of dimension ≤ 2 , however, this is not possible. That is birationally contractible proper surfaces are necessarily rational, atleast when k is algebraically closed of characteristics zero. In fact, any birtaionally connected surface is rational. Indeed, by the above corollary we know that such varieties are $\mathbb{A}^1$ -connected. But in characteristics zero such varieties are necessarily rational [AM11, Corollary 2.4.7].

Remark 3.6.20. The above proposition and its corollary are clearly false on non-proper spaces. Take, for example, $\mathbb{G}_m$. This space is $\mathbb{A}^1$ -disconnected despite being birationally connected (in fact, birationally contractible).

Remark 3.6.21. These might also fail for higher connectivity. For example, consider the proper motivic space $P := L_{mot}\mathbb{P}_k^2$ over a field k . Since $\mathbb{P}_k^2$ is rational over k , Corollary 2.2.9 implies $L_{bir}P \simeq *$. Therefore, $\pi_1^b P = *$, even though $\pi_1^{nis} P = \pi_1^{\mathbb{A}^1} \mathbb{P}^2 \cong \mathbb{G}_m$ [Mor12, Theorem 7.13].

We end this section with a birational analog of the Freudenthal Suspension theorem, which follows essentially from Theorem 3.6.6 and the methods of [Mor12, Theorem 6.61].

Theorem 3.6.22 (Birational Freudenthal suspension theorem). *Assume that S is Qcqs and that $\mathcal{E}$ is a pointed S -space that is birationally n -connected for some $n \geq 1$. Then:*

1. *The canonical adjoint unit map $\eta_{\mathcal{E}}^b : L_{bir}\mathcal{E} \rightarrow \Omega L_{bir}\Sigma\mathcal{E}$ has $2n$ -connected fiber.*
2. *The adjoint unit map $\eta_{\mathcal{E}} : \mathcal{E} \rightarrow \Omega\Sigma\mathcal{E}$ has a $2n$ -connected birational fiber. If $\mathcal{E}$ is connected, then, in fact, the ordinary fiber of $\eta_{\mathcal{E}}$ is birationally $2n$ -connected.*
3. *The canonical map $L_{bir}\mathcal{E} \rightarrow \Omega^\infty \Sigma_{bir}^\infty \mathcal{E}$ has $2n$ -connected fiber (here $\Sigma_{bir} := L_{bir}\Sigma$ is the internal suspension of $\mathcal{H}^b(S)$; equivalently, $\Sigma_{bir}\mathcal{X} = \text{colim}_{\mathcal{H}^b(S)}(pt \leftarrow L_{bir}\mathcal{X} \rightarrow pt)$).*

Proof. 1. We are given that $L_{bir}\mathcal{E}$ is n -connected. The classical Freudenthal suspension theorem, applied to this $2n$ -connected space $L_{bir}\mathcal{E}$, yields that the fiber of the unit $\eta_{L_{bir}\mathcal{E}} : L_{bir}\mathcal{E} \rightarrow \Omega\Sigma L_{bir}\mathcal{E}$ is $2n$ -connected. That is, the morphism

$$f : \pi_{2n+1}^b \mathcal{E} = \pi_{2n+1} L_{bir}\mathcal{E} \rightarrow \pi_{2n+2} \Sigma L_{bir}\mathcal{E}$$

is an epimorphism of presheaves, and

$$\tau_{\leq 2n} L_{bir}\mathcal{E} \xrightarrow{\tau_{\leq 2n} \eta_{L_{bir}\mathcal{E}}} \tau_{\leq 2n} \Omega\Sigma L_{bir}\mathcal{E} \simeq \Omega \tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E}$$

is an equivalence. Since the space $\tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E}$ is pointed and connected, this equivalence implies

$$\tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E} \simeq B\Omega \tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E} \simeq B\tau_{\leq 2n} L_{bir}\mathcal{E}.$$

Since $B\tau_{\leq 2n} L_{bir}\mathcal{E}$ is birational local by Lemma 3.3.4 and Lemma 3.2.3(2), it follows that so is $\tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E}$. But then

$$\tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E} \simeq L_{bir} \tau_{\leq 2n+1} \Sigma L_{bir}\mathcal{E} \simeq \tau_{\leq 2n+1} L_{bir} \Sigma L_{bir}\mathcal{E} \simeq \tau_{\leq 2n+1} L_{bir} \Sigma \mathcal{E}$$

(for the second equivalence use Proposition 3.6.3). Now, applying Ω to the above equivalence, we deduce that

$$\Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq \Omega\tau_{\leq 2n+1}L_{bir}\Sigma\mathcal{E} \simeq \tau_{\leq 2n}\Omega L_{bir}\Sigma\mathcal{E}.$$

Composing with the starting (given) equivalence $\tau_{\leq 2n}L_{bir}\mathcal{E} \simeq \Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$ this gives us an equivalence: $\tau_{\leq 2n}L_{bir}\mathcal{E} \simeq \tau_{\leq 2n}\Omega L_{bir}\Sigma\mathcal{E}$.

On the other hand, birational locality of $\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$ yields, by Theorem 3.6.6, that $\pi_{2n+2}\Sigma L_{bir}\mathcal{E} \rightarrow \pi_{2n+2}^b\Sigma L_{bir}\mathcal{E}$ is the universal one to a birational abelian group sheaf. But

$$\pi_i^b\Sigma L_{bir}\mathcal{E} = \pi_i L_{bir}\Sigma L_{bir}\mathcal{E} \simeq \pi_i L_{bir}\Sigma\mathcal{E} \simeq \pi_i^b\Sigma\mathcal{E}.$$

Therefore the morphism $\pi_{2n+2}\Sigma L_{bir}\mathcal{E} \rightarrow \pi_{2n+2}^b\Sigma\mathcal{E}$ is the universal morphism to a birational abelian group sheaf. Hence, the composition

$$\pi_{2n+1}^b\mathcal{E} \xrightarrow{f} \pi_{2n+2}\Sigma L_{bir}\mathcal{E} \rightarrow \pi_{2n+2}^b\Sigma\mathcal{E}$$

is an epimorphism in Ab_S^b (indeed, f is given to be an epimorphism of presheaves and thus an epimorphism on birationalization, while the second one is an isomorphism on birationalization). Since $Ab_S^b \subset PAb_S$ is exact (use Corollary 3.2.10 for the trivial topology), we deduce that $\pi_{2n+1}L_{bir}\mathcal{E} \rightarrow \pi_{2n+1}\Omega L_{bir}\Sigma\mathcal{E}$ is an epimorphism in PAb_S .

We have thus shown that the morphism $\eta_{\mathcal{E}}^b : L_{bir}\mathcal{E} \rightarrow \Omega L_{bir}\Sigma\mathcal{E}$ is an equivalence under $\tau_{\leq 2n}$ and induces a surjection on π_{2n+1} , which is equivalent to saying that it has $2n$ -connected fiber.

2. Since $\Sigma\mathcal{E}$ is connected, Theorem 3.2.12 implies that the codomain of the morphism in (1) is equivalent to $L_{bir}\Omega\Sigma\mathcal{E}$. Hence, the first statement of (2) is a restatement of (1). When $\mathcal{E}$ is furthermore connected, so that $\Omega\Sigma\mathcal{E}$ is connected, Lemma 3.2.11 implies that $L_{bir}fib(\eta_{\mathcal{E}}) \simeq fib(L_{bir}\eta_{\mathcal{E}})$, and the second line of (2) follows from the first.

3. The morphism in the third statement is the colimit of the following diagram:

$$L_{bir}\mathcal{E} \xrightarrow{\eta_{\mathcal{E}}} \Omega\Sigma_{bir}\mathcal{E} \xrightarrow{\Omega\eta_{\Sigma_{bir}\mathcal{E}}} \Omega^2\Sigma_{bir}^2\mathcal{E} \longrightarrow \dots$$

Inductively, it follows from statement (1) that each morphism in the above diagram is $2n$ connected. Hence, so is the colimit (use [Lur09, Proposition 6.5.1.12]). $\square$

4 Examples of Birational Motivic spaces

4.1 Birationally rigid schemes¹

Definition 4.1.1. *We say that a scheme T/k is birationally rigid if, for all dense open immersions $U \subset X$ of smooth schemes, the restriction maps $Sch_k(X, T) \rightarrow Sch_k(U, T)$ are isomorphisms, i.e., if it admits no nontrivial rational map.*

Example 4.1.2 (Abelian varieties). The standard example of a Birationally rigid variety is an abelian variety [Mil86, Theorem 3.2]. In particular, elliptic curves over k are birationally rigid.

¹This should not be confused with the notion of birational rigidity used in classical birational geometry.

Example 4.1.3 (Rationally rigid Proper varieties). Every rationally rigid proper variety (i.e., one with no rational curves) is birationally rigid. Indeed, suppose T is a proper variety over k having no rational curve, and we are given a morphism $f : U \rightarrow T$ with $U \in Sm_k$. Then by [Deb01, Corollary 1.44] f extends along any dense open immersion $U \hookrightarrow X \in Sm_k$. To elaborate, let us consider the graph of f , given by $\Gamma_f \subset U \times T \subset X \times T$. Let $\bar{\Gamma}_f \subset X \times T$ be the closure of Γ_f in $X \times T$, and let $\phi : \bar{\Gamma}_f \rightarrow X$ be the corresponding projection map, which is proper since T is so. Since $U \hookrightarrow X$ is dense and $U \cong \Gamma_f$, we get that ϕ is birational. Since X is smooth, the fibers of ϕ are rationally connected. Since T is rationally rigid, each fiber maps to a single point in T . Because ϕ is surjective by construction, we get an extension to all of X .

This applies in particular to $\mathbb{A}^1$ -rigid proper varieties and can be found in [Lev10, page 399].

4.2 Path components of standard correspondences

Let X be a smooth proper variety over a field k .

Example 4.2.1 (Chow groups of zero cycles of smooth proper varieties). The presheaf with transfers associated to the naive $\mathbb{A}^1$ -homotopy quotient of Y , i.e., $\pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{tr}(Y)$, is a birational local sheaf. Indeed, $\pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{tr}(Y) \cong CH_0(Y_{K-})$. So use [KS17, Theorem 2.1.2].

Example 4.2.2 (Chow witt groups of zero cycles of smooth proper varieties). Similarly, $\pi_0^{nis} L_{\mathbb{A}^1} \tilde{h}(Y)$, the group of Milnor-Witt points of a smooth proper variety Y , is birational local. One similarly proves that $\pi_0^{nis} L_{\mathbb{A}^1} \tilde{\mathbb{Z}}(Y) \cong \widetilde{CH}_0(Y_{K-})$. To see this, note that the second term is really a presheaf, that too birational and hence strongly $\mathbb{A}^1$ -invariant. The first term is also strongly $\mathbb{A}^1$ -invariant, being the zeroth homology sheaf of a Milnor-Witt motivic complex. The isomorphism follows from [Aso13, Corollary 4.3] after observing that, for finitely generated field extensions L/k , $\widetilde{Cor}(L, Y)/\widetilde{Cor}(\mathbb{A}_L^1, Y) \simeq \widetilde{CH}_0(Y_L)$.

Example 4.2.3 (Framed path components). Consider the moduli stack of framed points $h_k^{fr}(Y) : Sm_k^{op} \rightarrow Sets$ of a smooth proper variety Y . It turns out that $\pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{fr}(Y) \cong \widetilde{CH}_0(Y_{K-})$ (this follows from the main result of [Nes18]). The framed component $\pi_0^{nis} L_{\mathbb{A}^1} h_k^{fr}(Y)$ is thus birational.

4.3 More discrete birational sheaves

Example 4.3.1. Over geometrically unibranch schemes, every constant sheaf of sets is birational local. Indeed, in this case, the constant sheaf associated to a set F is given by $U \mapsto \sqcup_{U(0)} F$, since the connected components of unibranch schemes are irreducible.

Example 4.3.2. As noted earlier, for a field k , by [AM11, Theorem 6.1.7] any $\mathcal{F}_k^r$ – set that is sheaflike and transcendental invariant [Definition 6.1.6 of *loc.cit.*] can be extended to a birational sheaf on Sm_k .

4.4 Non-discrete birational local spaces

Example 4.4.1. It follows from Example 4.3.1 and Proposition 2.1.8 that over geometrically unibranch schemes, constant sheaves of spaces are birational local.

Example 4.4.2. For any n -birational motivic space $\mathcal{X}$ (see [Mai26c] for definition), the n -fold $\mathbb{P}^1$ -loop space $\Omega_{\mathbb{P}^1}^n \mathcal{X}$ is birational local [Mai26c, Corollary 4.5.2].

Example 4.4.3. Over a perfect field, the infinite loop space of a motivic $(S^1/\mathbb{P}^1)$ spectrum is birational-local if and only if the spectrum is zero-sliced (this is the main content of [Mai26d]).

Example 4.4.4 (Sifted colimits of birational local spaces). By Theorem 3.1.1, geometric realizations and filtered colimits of birational local spaces are birational local. For example, an Ind abelian variety is birational.

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Bibliography

- [ABH23] Aravind Asok, Tom Bachmann, and Michael J Hopkins, *On $\mathbb{P}^1$ stabilization in unstable motivic homotopy theory*, arXiv preprint arXiv:2306.04631 (2023).
- [AH11] Aravind Asok and Christian Haesemeyer, *Stable $\mathbb{A}^1$ -homotopy and R-equivalence*, J. Pure Appl. Algebra **215** (2011), no. 10, 2469–2472. MR 2793949
- [AM11] Aravind Asok and Fabien Morel, *Smooth varieties up to $\mathbb{A}^1$ -homotopy and algebraic h -cobordisms*, Adv. Math. **227** (2011), no. 5, 1990–2058. MR 2803793
- [Ann] Toni Annala, *Resolution of singularities*, <https://tannala.com/wp-content/uploads/2024/09/blowups.pdf>.
- [Aso13] Aravind Asok, *Birational invariants and $\mathbb{A}^1$ -connectedness*, J. Reine Angew. Math. **681** (2013), 39–64. MR 3181489
- [AWW17] Aravind Asok, Kirsten Wickelgren, and Ben Williams, *The simplicial suspension sequence in $\mathbb{A}^1$ -homotopy*, Geom. Topol. **21** (2017), no. 4, 2093–2160. MR 3654105
- [Ayo] Joseph Ayoub, *Counterexamples to f. morel’s conjecture on $\pi_0^{\mathbb{A}^1}$* .
- [Bac24] Tom Bachmann, *Strongly $\mathbb{A}^1$ -invariant sheaves (after F. Morel)*, arXiv preprint arXiv:2406.11526 (2024).
- [BE21] Tom Bachmann and Elden Elmanto, *Voevodsky’s slice conjectures via Hilbert schemes*, Algebr. Geom. **8** (2021), no. 5, 626–636. MR 4371542
- [BH21] Tom Bachmann and Marc Hoyois, *Norms in motivic homotopy theory*, Astérisque **425** (2021), ix+207. MR 4288071
- [BHS15] Chetan Balwe, Amit Hogadi, and Anand Sawant, *$\mathbb{A}^1$ -connected components of schemes*, Advances in Mathematics **282** (2015), 335–361.

- [BHS23] Chetan Balwe, Amit Hogadi, and Anand Sawant, *Strong $\mathbb{A}^1$ -invariance of $\mathbb{A}^1$ -connected components of reductive algebraic groups*, J. Topol. **16** (2023), no. 2, 634–649. MR 4637972
- [BPOsr22] Federico Binda, Doosung Park, and Paul Arne Østvær, *Motives and homotopy theory in logarithmic geometry*, C. R. Math. Acad. Sci. Paris **360** (2022), 717–727. MR 4449872
- [Cho14] Utsav Choudhury, *Connectivity of motivic H -spaces*, Algebr. Geom. Topol. **14** (2014), no. 1, 37–55. MR 3158752
- [CK23] Denis-Charles Cisinski and Bruno Kahn, *Homotopy theory of schemes and r -equivalence*, arXiv preprint arXiv:2311.00092 (2023).
- [CM21] Dustin Clausen and Akhil Mathew, *Hyperdescent and étale K -theory*, Invent. Math. **225** (2021), no. 3, 981–1076. MR 4296353
- [CR24] Utsav Choudhury and Biman Roy, *$\mathbb{A}^1$ -connected components and characterisation of $\mathbb{A}^2$* , J. Reine Angew. Math. **807** (2024), 55–80. MR 4698492
- [Deb01] Olivier Debarre, *Higher-dimensional algebraic geometry*, Universitext, Springer-Verlag, New York, 2001. MR 1841091
- [EHK⁺21] Elden Elmanto, Marc Hoyois, Adeel A Khan, Vladimir Sosnilo, and Maria Yakerson, *Motivic infinite loop spaces*, Cambridge Journal of Mathematics **9** (2021), no. 2, 431–549.
- [Gro66] Alexander Grothendieck, *Éléments de géométrie algébrique: Iv. étude locale des schémas et des morphismes de schémas, troisième partie*, Publications Mathématiques de l’IHÉS **28** (1966), 5–255.
- [Har77] Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. No. 52, Springer-Verlag, New York-Heidelberg, 1977. MR 463157
- [Hoy14] Marc Hoyois, *A quadratic refinement of the Grothendieck-Lefschetz-Verdier trace formula*, Algebr. Geom. Topol. **14** (2014), no. 6, 3603–3658. MR 3302973
- [Hoy17] ———, *The six operations in equivariant motivic homotopy theory*, Adv. Math. **305** (2017), 197–279. MR 3570135
- [Koi22] Junnosuke Koizumi, *Zeroth $\mathbb{A}^1$ -homology of smooth proper varieties*, New York J. Math. **28** (2022), 824–834. MR 4436127
- [KS15] Bruno Kahn and R. Sujatha, *Birational geometry and localisation of categories*, Doc. Math. (2015), 277–334, With appendices by Jean-Louis Colliot-Thélène and Ofer Gabber. MR 3404383
- [KS17] Bruno Kahn and Ramdorai Sujatha, *Birational motives, II: Triangulated birational motives*, Int. Math. Res. Not. IMRN **22** (2017), 6778–6831. MR 3737321
- [Lev10] Marc Levine, *Slices and transfers*, Doc. Math. (2010), 393–443. MR 2804260

- [Lur09] Jacob Lurie, *Higher topos theory*, Princeton University Press, 2009.
- [Mai26a] Dipankar Maity, $\mathbb{P}^1$ *homotopy theory*, In preparation (2026).
- [Mai26b] ———, *On the internal homotopy theory of motivic categories*, In preparation (2026).
- [Mai26c] ———, *Schematic functorialities of the unstable n birational motivic homotopy categories*, In preparation (2026).
- [Mai26d] ———, *Zero slices through unstable birationality*, In preparation (2026).
- [Mil80] J. S. Milne, *étale cohomology*, Princeton Mathematical Series, vol. 33, Princeton University Press, Princeton, NJ, 2025] ©1980, Reprint of [0559531]. MR 4904233
- [Mil86] ———, *Abelian varieties*, Springer, New York, 1986. MR 861974
- [Mor12] Fabien Morel, *$\mathbb{A}^1$ -algebraic topology over a field*, Lecture Notes in Mathematics, vol. 2052, Springer, Heidelberg, 2012. MR 2934577
- [MV99] Fabien Morel and Vladimir Voevodsky, *$\mathbb{A}^1$ -homotopy theory of schemes*, Inst. Hautes Études Sci. Publ. Math. **90** (1999), 45–143. MR 1813224
- [Nes18] Alexander Neshitov, *Framed correspondences and the Milnor-Witt K -theory*, J. Inst. Math. Jussieu **17** (2018), no. 4, 823–852. MR 3835524
- [Pel14] Pablo Pelaez, *The unstable slice filtration*, Trans. Amer. Math. Soc. **366** (2014), no. 11, 5991–6025. MR 3256191
- [Rio02] Joël Riou, *Théorie homotopique des s -schémas*, Memoire de DEA realise sous la direction de Bruno Kahn (2002).
- [Shi23] Yuri Shimizu, *Universal birational invariants and $\mathbb{A}^1$ -homology*, Comm. Algebra **51** (2023), no. 2, 807–821. MR 4532827
- [Sta26] The Stacks project, *The stacks project*, <https://stacks.math.columbia.edu>, 2026.